

From Predictive Dynamics to Spectral Geometry

Aurélien Pélissier[†]

[†]Correspondence: aurelien.pelissier.38@gmail.com

Abstract

In Alain Connes’ noncommutative geometry, geometry is encoded by an algebra, a Hilbert space, and a Dirac operator taken as fundamental spectral data. We show that these structures can instead emerge from a more primitive predictive dynamics. A compatible first-order variation on the induced history algebra canonically generates a real even Hodge–Dirac spectral triple, while the algebra alone determines neither its metric scale nor the full Dirac operator. The resulting spectral geometry is an information-reduced description: dynamically inequivalent predictive systems can share the same effective spectral data. Nevertheless, finite geometries become reconstructible after finitely many calibrations, and on periodic three-dimensional lattice approximants the emergent spectral distance converges to a Riemannian metric reconstructed independently from the dynamics. Thus Connes-type spectral geometry may arise not as fundamental structure, but as an effective geometric description of richer underlying dynamics. This provides a concrete setting in which noncommutative geometry need not be fundamental, but may instead arise as a coarse-grained spectral description. A `Lean` formalization is available at <https://github.com/AI-MathPhys/renewal-geometry>.

Keywords. Noncommutative geometry; spectral triples; Renewal Geometry; predictive dynamics; Hodge–Dirac operators; Connes distance; emergent geometry; spectral reconstruction.

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1 Introduction

Many spaces that arise naturally in mathematics and physics do not admit a satisfactory description as ordinary geometric spaces. Quotients may be singular, points may fail to capture the relevant observables, and the structures of interest may be encoded more faithfully by operators than by a classical manifold. In such situations the usual geometric language—built from points, coordinates, and functions on a well-behaved space—can cease to be adequate, even though an algebra of observables remains meaningful.

Noncommutative geometry was developed to address this problem. Connes’ programme replaces the requirement of an underlying point-set space by an operator-algebraic description in which an algebra plays the role of generalized coordinates. In the commutative case this remains compatible with ordinary geometry, since a suitable commutative algebra determines the underlying space through Gelfand duality. Allowing the algebra to become noncommutative extends the same principle to settings for which no classical space of points need exist.

The spectral formulation equips this generalized coordinate algebra with geometry. A compact spin manifold can be encoded by an algebra $\mathcal{A}$ represented on a Hilbert space $\mathcal{H}$, together with a Dirac operator D [1]. The representation supplies the operator-theoretic carrier, while D encodes differential and metric information. Under the reconstruction axioms for commutative spectral triples, this operator data recovers compact spin geometry [2, 3]; in the genuinely noncommutative case the same operator-geometric language continues to apply even when no ordinary underlying space exists.

Spectral triples consequently provide a common language for noncommutative spaces, differential calculi, index theory, and metric geometry. The Connes distance formula turns the commutator with D into a metric notion, making the Dirac operator simultaneously a differential and geometric object. Related work on compact quantum metric spaces formulates metric information at the level of an algebra equipped with a Lipschitz seminorm [4]. Rieffel’s quantum Gromov–Hausdorff distance and the later propinquity and spectral propinquity provide frameworks for convergence of quantum metric spaces and metric spectral triples [5, 6, 7], while real structures and gradings encode analogues of spin and orientation data.

A different structural question is whether the objects entering this spectral description must themselves be regarded as primitive. This is logically prior to spectral reconstruction: rather than asking which geometry is encoded by a given spectral triple, one asks whether the algebra, representation, differential structure, and Dirac data can arise from an underlying dynamics in which none of them is initially prescribed.

From predictive dynamics to spectral geometry. Renewal Geometry provides one setting in which this question can be posed operationally [8]. Its starting point is not a manifold, graph, causal order, algebra, Hilbert space, or spectral triple, but a finite operational process specified by composable intervention histories and their conditional future statistics. Histories that cannot be distinguished by any admissible future continuation are identified. The resulting future-equivalence quotient reconstructs the minimum persistent predictive carrier of the process rather than requiring such a carrier to be supplied independently.

Complete operational statistics then reconstruct a canonical positive history object, the *Renewal Grand Tensor*. We denote the Grand Tensor associated with a finite predictive system X by $\mathbb{T}_{\text{Grand}, X}$. Its role is pregeometric: it records which distinctions survive prediction, how represented histories compose, and how they overlap, before positions, distances, metrics, or continuum geometry have been assigned. The corresponding history algebra and positive moment structure therefore

arise from the operational process rather than being inserted as geometric input.

For the present paper, the relevant interface is the algebraic content carried by $\mathbb{T}_{\text{Grand},X}$. The Grand Tensor canonically supplies the future-minimal predictive carrier, the represented history algebra $\mathcal{A}_X^{\text{hist}}$, and its positive moment structure. On the finite algebra there is in particular an intrinsic normalized regular trace, which will provide one canonical choice of τ_X . To pass from this reconstructed algebraic object to spectral geometry, one then specifies a compatible first-order variation δ_X of the history algebra. Thus the interface used here is

$$\mathbb{T}_{\text{Grand},X} \longmapsto (\mathcal{A}_X^{\text{hist}}, \tau_X) \xrightarrow{\delta_X} (\mathcal{A}_X, \mathcal{H}_X, D_X, J_X, \gamma_X). \quad (1)$$

The first arrow is reconstructed directly from the operational statistics and is recalled self-containedly in Section 2. The second is the Hodge–Dirac spectralization studied in this paper. The Grand Tensor is not identified with the displayed algebraic or spectral data: it retains the predictive carrier and complete positive history structure, including information that need not survive in the spectral image. Nor do we assume that the Grand Tensor or its algebra alone fixes δ_X ; the first-order variation is exactly the additional datum whose metric scale and Dirac ambiguity are analyzed below.

The same $\mathbb{T}_{\text{Grand},X}$ also supports the separate Lorentzian spacetime construction developed within the Renewal Geometry framework [8]. The present work asks instead whether Connes-type spectral geometry can arise from a compatible first-order variation of its reconstructed history algebra. The spacetime-specific assumptions of the companion construction are not used below; once a finite-dimensional unital C^* -algebra $\mathcal{A}$, a faithful tracial state τ , and a compatible real Hilbert-bimodule derivation δ are supplied, the finite spectralization results are ordinary operator-theoretic statements independent of the operational route by which $\mathcal{A}$ was obtained.

The resulting problem has both a forward and an inverse aspect. On the forward side one asks whether a first-order predictive differential canonically produces the structures of spectral geometry, and whether the resulting metric has an independent dynamical meaning. On the inverse side one asks which spectral geometries can be reached, what additional data are required to reconstruct a full Dirac operator, and whether different microscopic dynamics can have the same spectral image.

What lies before a spectral triple? At the finite-dimensional level, Krajewski classified finite spectral triples in terms of finite matrix data and diagrams [9]. This gives a powerful classification of already specified finite spectral geometries, but it does not address whether the algebraic carrier or Dirac data can themselves be reconstructed from an operational dynamics.

Several constructions derive spectral triples from other supplied geometric or hierarchical structure. For approximately finite-dimensional C^* -algebras, a Dirac operator can be built naturally from a prescribed finite-dimensional filtration [10]. Weighted trees, ultrametric Cantor sets, fractal spaces, stationary Bratteli diagrams, and aperiodic systems provide further constructions in which the underlying space, algebra, hierarchy, or incidence structure is already given [11, 12, 13]. These results show that the Dirac operator need not be specified independently, but their mathematical entrance remains a pre-existing algebraic or geometric carrier.

A related dynamical literature constructs or extends spectral triples from a prescribed C^* -dynamical system. Crossed-product constructions attach spectral data to given group actions [14]; for symbolic dynamics, spectral triples have been constructed on subshift algebras and their crossed products [15]; and analogous constructions have been developed for irreversible C^* -dynamical systems [16]. Here again the dynamics acts on an algebra that is supplied before spectralization. The present question differs at the entrance: the history algebra itself is reconstructed only after predictive future-equivalence has removed operationally invisible refinements.

A particularly close line of work comes from quantum Riemannian geometry. Majid has shown that a geometrically realized Dirac operator can be constructed canonically from a prescribed noncommutative coordinate algebra, differential calculus, quantum metric, and compatible bimodule connection, with explicit realizations on graphs, lattices, the fuzzy sphere, and matrix algebras [17]. This places the Dirac operator downstream of quantum metric and differential data. The present question begins one level earlier: neither the coordinate algebra nor the differential and metric structures from which the Dirac operator is built are assumed at the operational entrance.

An even closer dynamical precedent is provided by noncommutative potential theory. Cipriani and Sauvageot showed that a tracially symmetric Dirichlet form on a C^* -algebra admits a canonical realization through a closable derivation into a Hilbert bimodule, with the Markov generator represented by the corresponding divergence [18]. This calculus has been used to define Dirac operators on fractals [19], and Hinz, Kelleher and Teplyaev constructed spectral triples from regular Dirichlet and resistance forms, showing in the compact strongly local case that the Connes distance coincides with the intrinsic metric [20]. Subsequent work has related Hodge–Dirac spectral triples of sub-Markovian semigroups to heat-kernel and spectral-dimension estimates [21, 22]. These constructions are especially close to the Hodge–Dirac layer used below, but they begin with the algebra, reference state or trace, and Dirichlet or Markov structure already supplied.

The operational entrance also has established precedents. Quantum combs and process-tensor formalisms encode complete multitime intervention statistics by positive causal process objects and describe memory-bearing quantum dynamics without imposing a Markov approximation [23, 24, 25]. They reconstruct or represent a multitime process from intervention statistics, but do not by themselves identify histories by predictive future-equivalence and then use that quotient to construct the coordinate algebra of a spectral geometry.

There is also a broad classical literature in which state is defined through future predictions rather than hidden microscopic labels. Observable operator models encode stochastic processes through operators acting on observable statistics [26], while predictive-state representations use multi-step, action-conditional predictions as state coordinates [27, 28]. In computational mechanics, causal states identify pasts that induce the same conditional distribution over futures and yield minimal predictive presentations [29, 30, 31]; continuous-time extensions include renewal and discrete-event processes [32]. Renewal Geometry combines this predictive quotient with multitime operational composition and positive process structure. The persistent carrier and represented history algebra are reconstructed only after future-invisible distinctions have been removed, and the first-order differential is then spectralized on that reconstructed algebra.

The question considered here can therefore be stated as follows:

Can the algebra, Hilbert-space carrier, differential structure, and Dirac data of noncommutative geometry arise from an underlying predictive dynamics in which none of them is initially prescribed?

A nontrivial answer requires more than reproducing a desired spectral triple by definition. The algebra cannot simply be hidden in the microscopic model under another name, and the Dirac operator cannot be fitted afterward to recover a target metric. One must determine what the differential actually fixes, which spectral geometries are obtainable under different notions of reconstruction, whether the resulting metric can be validated independently, and what information is lost in the passage from dynamics to spectral geometry.

The last point is central. If distinct predictive systems have the same spectral image, then a spectral triple is not, in general, a complete invariant of the dynamics from which it arose. It may instead describe an equivalence class of microscopically different systems that agree on their

effective metric and differential structure. In this sense spectral geometry can be *information-reduced*: complete for the operator-geometric information it encodes while insufficient, in general, to reconstruct the richer underlying dynamics.

Our contribution. We first construct a canonical spectralization of finite differential data. A star-compatible derivation δ into a real Hilbert $\mathcal{A}$ -bimodule determines a minimal one-form carrier and a real even Hodge–Dirac spectral triple with exact order-zero and first-order relations. Two rigidity results identify what this construction does not determine. The algebra and faithful trace do not fix an absolute Dirac scale, and a fixed commutator derivation determines the full compatible Dirac operator only modulo a finite-dimensional self-adjoint commutant. The orthogonally centered representative is the unique minimum-Hilbert–Schmidt-norm implementer, and the dimension of this commutant is the sharp number of independent scalar measurements required to distinguish the remaining Dirac ambiguity.

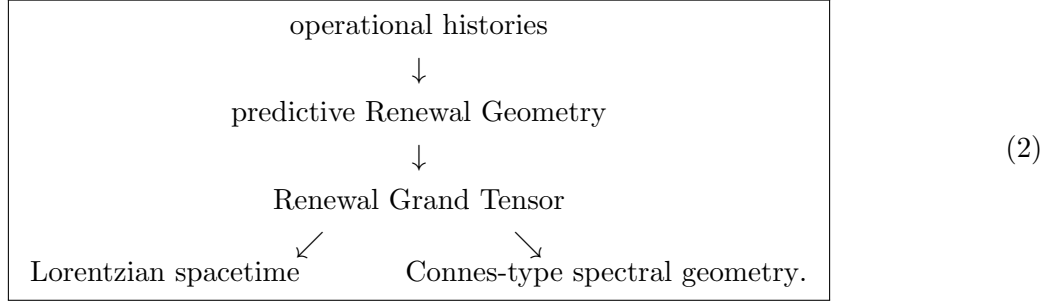
This distinction separates an *austere Hodge construction* from an enlarged *calibrated marked reconstruction*. The bare Hodge operator always has the unit zero-mode and therefore cannot realize arbitrary finite Dirac operators. When the missing finite calibration data are retained, every finite spectral packet is reconstructible and every separable compact-resolvent packet arises as a strong cofinal limit of finite marked reconstructions. A stricter requirement, in which multiplication must be transported asymptotically in norm, is genuinely selective and is characterized exactly by spectral quasidiagonality.

We next test whether the construction has independent geometric content. For stationary graph dynamics, the symmetric jump data determine a Hodge–Dirac operator whose square is the symmetric Markov generator and whose commutator seminorm is its local quadratic form. On periodic A_3 approximants, the corresponding Connes distance converges uniformly to a Riemannian metric reconstructed independently from the jump second moments. For closed Riemannian spin manifolds, the local positive-stencil and covariant Wilson construction gives the required chartwise estimates, while global norm-resolvent convergence follows under the stated compatible stable-spin discretization hypotheses. The independent scalar metric reconstruction is unconditional.

Finally, we characterize several mechanisms by which spectralization loses information. Directed stationary current can vary at fixed metric spectral geometry, hidden dynamical memory can vary at fixed static spectral data, and locally identical spectral packets can differ by global marked monodromy. These inverse fibres have intrinsic coordinates and canonical coarse-graining maps. Thus spectralization is not merely noninjective: the information it forgets has its own mathematical structure and transformation laws.

The resulting interpretation is that Connes-type spectral geometry can arise as an effective geometric description of a richer predictive dynamics rather than necessarily as its most primitive level. This conclusion is strengthened by the companion spacetime construction: Lorentzian spacetime and spectral geometry can both arise downstream of the same pregeometric predictive structure, without either being required as the primitive entrance to the other. The two papers

therefore lead to the schematic picture



The left branch is developed in the companion Renewal Geometry work [8]. The present paper develops the right branch: its canonical finite construction, its category-dependent image, its independent metric test, and its inverse fibres. The central question is therefore not only why spectral geometry can arise, but what a richer underlying dynamics retains that must be forgotten when it is compressed to a spectral description.

2 From predictive dynamics to spectral geometry

The spectralization theorem is algebraically self-contained, but its motivating entrance is more primitive than the datum $(\mathcal{A}, \tau, \delta)$. We therefore begin with the finite predictive construction of the Renewal Grand Tensor [8], restated here in the finite form needed for the present paper. This makes explicit which structures are reconstructed before spectral geometry is introduced and which first-order information enters only at the spectralization stage.

2.1 The Renewal Grand Tensor and the first-order differential interface

Fix a finite cutoff X . The operational entrance consists of a finite typed event grammar with finite reachable histories. If a history h ends at cut type x , let $\mathbf{F}_{X,x}$ be the finite admissible future words beginning at x . The conditional event law assigns normalized one-step probabilities and therefore cylinder probabilities

$$p_X(w \mid h) = \prod_{j=1}^k p_X(a_j \mid ha_1 \cdots a_{j-1}), \quad w = a_1 \cdots a_k, \quad (3)$$

with $p_X(\epsilon_x \mid h) = 1$. No persistent state space, algebra, Hilbert space, metric, or Dirac operator is assumed at this stage.

Two histories of the same cut type are *future equivalent* when no declared finite continuation distinguishes them:

$$h \sim_X h' \iff p_X(w \mid h) = p_X(w \mid h') \text{ for every } w \in \mathbf{F}_{X,x}. \quad (4)$$

Histories of different cut type are never identified. This is a typed right congruence. Indeed, if $h \sim_X h'$ and $a : x \rightarrow y$ is admissible, then the one-letter future gives $p_X(a \mid h) = p_X(a \mid h')$; on a positive-probability branch, for every continuation v ,

$$p_X(v \mid ha) = \frac{p_X(av \mid h)}{p_X(a \mid h)} = \frac{p_X(av \mid h')}{p_X(a \mid h')} = p_X(v \mid h'a).$$

Hence $ha \sim_X h'a$.

Define the minimum persistent predictive carrier by

$$\boxed{Z_{X,x}^{\min} := H_{X,x} / \sim_X, \quad q_{X,x} : H_{X,x} \rightarrow Z_{X,x}^{\min}} \quad (5)$$

For an admissible event $a : x \rightarrow y$, right congruence gives canonical update and branch maps

$$\boxed{T_{X,a}([h]) = [ha], \quad \kappa_{X,a}([h]) = p_X(a \mid h).} \quad (6)$$

Thus persistence is reconstructed from the conditional future law rather than introduced as an independent coordinate. Moreover, this quotient is universal: if S_X is any reachable deterministic predictive realization of the same conditional event law, then equality of two realization states forces equality of all their future cylinder probabilities. Consequently there is a unique surjection

$$\pi_{X,x} : S_{X,x} \rightarrow Z_{X,x}^{\min} \quad (7)$$

intertwining branch probabilities and updates. Hence $Z_X^{\min}$ is the unique coarsest reachable deterministic predictive realization.

To reconstruct the operator-algebraic history object, assume in addition that the finite multitime intervention tables are specified on tomographically complete finite operator frames, are affine under randomization and additive under coarse graining, and obey the usual finite positivity, normalization, prefix compatibility, and causal-discard conditions. Finite tomography then determines the corresponding positive causal process matrices. Choosing their support-minimal square-root realizations introduces no additional invariant: minimal realizations of the same positive process data differ only by unitaries on the support spaces. The represented predictive-state projections, mixed forward maps, and their Hilbert–Schmidt reverses therefore generate, up to represented $*$ -isomorphism, a finite history algebra

$$\mathcal{A}_X^{\text{hist}} \subseteq \mathcal{B}(\mathcal{M}_X). \quad (8)$$

Here $\mathcal{M}_X$ denotes any support-minimal sequential carrier; its particular choice is immaterial up to the preceding unitary equivalence.

On every finite-dimensional C^* -algebra there is a canonical normalized regular trace. For $\mathcal{A}_X^{\text{hist}} \cong \bigoplus_{\lambda} M_{n_{\lambda}}(\mathbb{C})$, set

$$\widehat{\tau}_{X,\text{reg}}(a) = \frac{\text{Tr}_{\mathbb{C}}(L_a)}{\dim_{\mathbb{C}} \mathcal{A}_X^{\text{hist}}} = \frac{\sum_{\lambda} n_{\lambda} \text{Tr}(a_{\lambda})}{\sum_{\lambda} n_{\lambda}^2}. \quad (9)$$

If $[w]$ denotes the represented operator associated with a finite history word, define the positive word-Gram kernels

$$\mathbb{K}_{X,r}(v, w) = \widehat{\tau}_{X,\text{reg}}([v]^*[w]), \quad |v|, |w| \leq r. \quad (10)$$

Positivity is immediate from the trace inner product. Since $\mathcal{A}_X^{\text{hist}}$ is finite dimensional, the increasing spans of represented words stabilize after finite depth; once this happens, one additional word layer determines multiplication and involution on the stabilized span.

Definition 2.1 (Finite Renewal Grand Tensor). The finite Renewal Grand Tensor associated with the operational process X is

$$\boxed{\mathbb{T}_{\text{Grand},X} = \left(\mathcal{A}_X^{\text{hist}}, \{ \mathbb{K}_{X,r} \}_{r \geq 0}, Z_X^{\min} \right).} \quad (11)$$

It is a tensor in the operational sense of a complete positive history-moment object: $Z_X^{\min}$ records which past distinctions remain predictively effective, $\mathcal{A}_X^{\text{hist}}$ records how represented histories compose, and the Gram hierarchy records their intrinsic positive overlaps.

Proposition 2.2 (Canonical finite Grand-Tensor reconstruction). *Under the finite assumptions above, the following data are fixed by the complete operational statistics up to their natural isomorphisms:*

- (i) *the future-minimal carrier $Z_X^{\min}$ and its update/branch maps;*
- (ii) *the support-minimal represented history algebra $\mathcal{A}_X^{\text{hist}}$, up to represented $*$ -isomorphism;*
- (iii) *the normalized regular trace and the positive word-Gram hierarchy;*
- (iv) *consequently, the Grand Tensor $\mathbb{T}_{\text{Grand},X}$ of (11).*

No independent hidden-state presentation, persistent state space, metric, or spectral triple is needed for this reconstruction.

Proof. Right congruence was proved above. The universal factorization (7) shows that $Z_X^{\min}$ is independent of a chosen predictive presentation. Tomographic completeness makes the finite multitime positive process matrices unique; support-minimal Gram factorizations of the same positive matrices are unitarily equivalent. The represented operators generated from such a realization therefore determine the same finite C^* -algebra up to represented $*$ -isomorphism. The normalized regular trace is intrinsic to that algebra, so the Gram kernels (10) are invariant under the same equivalence. Finite dimensionality gives stabilization of the represented word spans, completing the claim. $\square$

This proposition is the self-contained predictive entrance needed in the present paper. The same construction is developed in the broader spacetime setting of the Renewal Geometry framework [8], including additional same-history physical marks needed by the Lorentzian branch; none of those spacetime-specific marks is assumed for the spectralization below.

First-order differential interface. The Grand Tensor is logically upstream of spectral geometry, but it is important not to identify algebraic reconstruction with metric differentiation. From (11) we canonically obtain the history algebra and its positive finite-dimensional structure. We take

$$\mathcal{A} = \mathcal{A}_X^{\text{hist}}, \quad \tau = \hat{\tau}_{X,\text{reg}}, \quad (12)$$

or, more generally, any declared faithful tracial state on $\mathcal{A}$. To spectralize this algebra one additionally specifies a compatible first-order variation. Abstractly, this is a Hilbert $\mathcal{A}$ -bimodule $\mathcal{K}$, an antiunitary involution $C_{\mathcal{K}}$ reversing the bimodule actions, and a complex-linear map

$$\delta : \mathcal{A} \longrightarrow \mathcal{K} \quad (13)$$

satisfying

$$\delta(\mathbf{1}) = 0, \quad \delta(ab) = a\delta(b) + \delta(a)b, \quad C_{\mathcal{K}}\delta(a) = \delta(a^*). \quad (14)$$

When this datum comes from a Renewal process, δ represents the declared first-order variation of the reconstructed predictive history algebra. The results below do not assume that $\mathbb{T}_{\text{Grand},X}$ or $\mathcal{A}$ alone uniquely determines this variation. This distinction is essential: the no-scale result below shows that rescaling δ changes the spectral metric while leaving the algebra and trace fixed.

The logical interface is therefore

$$\boxed{\text{operational histories} \longrightarrow \mathbb{T}_{\text{Grand},X} \longmapsto (\mathcal{A}_X^{\text{hist}}, \tau_X) \xrightarrow{\delta_X} (\mathcal{A}_X, \mathcal{H}_X, D_X, J_X, \gamma_X).} \quad (15)$$

The first arrows reconstruct predictive and algebraic structure; the final arrow is the Hodge–Dirac spectralization studied in this paper.

The zero-form carrier is

$$\mathcal{H}_0 = L^2(\mathcal{A}, \tau), \quad (16)$$

and the one-form carrier is the minimal quotient of the universal calculus generated by δ ,

$$\mathcal{H}_1(\delta) = \Omega_u^1(\mathcal{A}) / \text{Ker } T_\delta, \quad \Omega_u^1(\mathcal{A}) = \text{Ker}(m : \mathcal{A} \otimes \mathcal{A} \rightarrow \mathcal{A}). \quad (17)$$

On $\mathcal{H} = \mathcal{H}_0 \oplus \mathcal{H}_1(\delta)$, the induced derivation ∂ gives

$$D = \begin{pmatrix} 0 & \partial^* \\ \partial & 0 \end{pmatrix}, \quad \gamma = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad (18)$$

while reversal of universal one-forms gives the real structure J . Thus the spectral triple is constructed from a first-order differential on an algebra that has itself already been reconstructed from predictive histories.

2.2 What the differential determines

The finite Hodge construction is canonical once δ is fixed. It produces a real even spectral triple with exact order-zero and first-order relations. The spectral scale is not already hidden in $(\mathcal{A}, \tau)$: replacing δ by $s\delta$ leaves the algebra and trace unchanged but sends

$$D \mapsto sD, \quad L_D \mapsto sL_D, \quad d_D \mapsto s^{-1}d_D. \quad (19)$$

Thus an absolute metric scale enters through the differential data.

There is a second, finite ambiguity. For a marked spectral packet $\mathbf{S} = (\mathcal{A}, \mathcal{H}, \pi, D, J, \gamma)$ define

$$\mathfrak{c}_{\mathbf{S}} = \{C \in \pi(\mathcal{A})'_{\text{sa}} : C\gamma + \gamma C = 0, JC = \varepsilon' C J\}. \quad (20)$$

All compatible self-adjoint Dirac operators implementing the same commutator derivation form exactly the affine space

$$D + \mathfrak{c}_{\mathbf{S}}. \quad (21)$$

The orthogonally centered representative $D_{\text{der}} = D - P_{\mathfrak{c}_{\mathbf{S}}} D$ is the unique minimum-Hilbert–Schmidt-norm implementer, and $q_{\mathbf{S}} = \dim_{\mathbb{R}} \mathfrak{c}_{\mathbf{S}}$ real linear scalar measurements are necessary and sufficient to distinguish every point of the fibre.

The austere Hodge construction has an additional visible restriction: $D(\mathbf{1}, 0) = 0$. It therefore cannot be surjective onto arbitrary finite Dirac operators. This is why the inverse statement below uses an enlarged calibrated marked reconstruction rather than the bare Hodge functor.

2.3 Which spectral geometries can be reconstructed?

At the finite marked level, one is allowed to retain finitely many calibration data, including a target generator germ or the equivalent commutant coordinates. At the strong compact level, finite spectral compressions converge strongly on fixed vectors, resolvents, commutators, and matrix coefficients. At the strict norm-monoidal level, multiplication must additionally be transported asymptotically in norm.

Theorem 2.3 (Spectralization, Dirac ambiguity, and calibrated image). *The following statements hold.*

- (i) *Every finite real differential datum $(\mathcal{A}, \tau, \delta)$ determines the real even Hodge–Dirac spectral triple above, functorially under trace- and differential-preserving $*$ -isomorphisms.*
- (ii) *The algebra and trace do not determine the Dirac scale, and a fixed commutator derivation determines the full compatible Dirac operator exactly modulo $\mathfrak{c}_S$.*
- (iii) *For the enlarged calibrated marked reconstruction,*

$$\begin{aligned}
\text{EssIm}(\mathfrak{S}_{\text{fin}}^{\text{mark}}) &= \mathbf{Spec}_{\text{fin}}, \\
\text{EssIm}(\mathfrak{S}_{\text{cpt}}^{\text{strong,mark}}) &= \mathbf{Spec}_{\text{cpt}}^{\text{sep}}, \\
\text{EssIm}(\mathfrak{S}_{\text{cpt}}^{\text{strict}}) &= \mathbf{Spec}_{\text{cpt}}^{\text{sqd}} \subsetneq \mathbf{Spec}_{\text{cpt}}^{\text{sep}}.
\end{aligned} \tag{22}$$

Thus every finite spectral packet is reconstructible once the finite calibration marks are retained; every separable compact-resolvent packet is a strong cofinal limit of such finite marked reconstructions; and strict norm-monoidal realizability is equivalent to spectral quasidiagonality.

The distinction between the first and third clauses is essential. Finite marked surjectivity is a reconstruction benchmark for an enlarged data set; it is not a claim that an arbitrary target D has been derived from the algebra or Hodge derivation alone. The complete definitions and proofs are given in [Section S1](#) of the mathematical supplement.

3 Independent metric reconstruction and compact spin geometry

Producing an operator called D is not by itself evidence that geometry has emerged. A stronger test is available when the same finite dynamics determines a metric by a route that does not use the Dirac operator.

3.1 An independent metric test

For a stationary finite graph, let $m_x > 0$ be the stationary masses, $q_{xy} = m_x k_{xy}$ the stationary fluxes, and

$$c_{xy} = \frac{q_{xy} + q_{yx}}{2}, \quad j_{xy} = \frac{q_{xy} - q_{yx}}{2}. \tag{23}$$

The symmetric conductances determine the graph Hodge–Dirac operator $D_{m,c}$. Exactly,

$$D_{m,c}^2|_{\ell^2(V,m)} = -2\mathcal{L}_s, \quad \|[D_{m,c}, f]\|^2 = \max_x \frac{1}{m_x} \sum_y c_{xy} |f(y) - f(x)|^2. \tag{24}$$

Thus the Dirac square is the symmetric Markov generator and its commutator seminorm is the local quadratic form of that generator.

On the periodic A_3 approximants, the normalization gives

$$L_h(f)^2 = \max_{x \in V_h} \frac{1}{8h^2} \sum_{\alpha \in \Phi_{A_3}} |f(x + h\alpha) - f(x)|^2, \tag{25}$$

and the pure-state Connes distance satisfies

$$\sup_{x,y \in V_h} |d_h^{\text{Connes}}(x, y) - d_g(x, y)| \longrightarrow 0, \tag{26}$$

where d_g is reconstructed independently from the jump second moments. This is the principal anti-circularity test of the paper.

3.2 From local spin discretization to a global convergence criterion

The spin problem is subtler. Local positive stencils reconstruct the inverse metric, and a density-symmetric covariant Wilson discretization removes lattice doublers while remaining $O(h)$ on sampled smooth spinors. These chartwise estimates do not by themselves guarantee that independently discretized charts assemble into one stable global operator. The mathematical supplement therefore isolates the additional global approximation properties in the notion of a *compatible stable spin discretization*.

Theorem 3.1 (Independent metric reconstruction and stable compact-spin convergence). *The following statements hold.*

- (i) *The periodic A_3 Hodge–Dirac operator obeys (24), and its Connes distance converges uniformly as in (26).*
- (ii) *The relation-adapted cellular completion removes the extensive one-form kernel and has the expected compact Hodge–Dirac continuum limit.*
- (iii) *The covariant Wilson construction has one low-energy species and satisfies the local uniform graph estimates required for a spin continuum limit.*
- (iv) *Let $(M^d, g, \mathfrak{s})$ be closed. If the finite covariant Wilson operators form a compatible stable spin discretization in the precise sense of Definition S4.7, then for every $z \in \mathbb{C} \setminus \mathbb{R}$,*

$$\left\| W_h(D_h - z)^{-1} W_h^* - (\widehat{D}_{M, \mathfrak{s}} - z)^{-1} \right\| \longrightarrow 0, \quad \widehat{D}_{M, \mathfrak{s}} = D_{M, \mathfrak{s}} \otimes \sigma_1. \quad (27)$$

- (v) *Independently of the spin-assembly hypothesis, every closed Riemannian manifold admits finite metric triples on refining meshes for which*

$$\sup_{x, y \in V_h} |d_h^{\text{met}}(x, y) - d_g(x, y)| \longrightarrow 0. \quad (28)$$

The theorem deliberately separates the verified local spin analysis from the global atlas-stability requirement. The scalar metric approximation is unconditional; the spin norm-resolvent conclusion is a convergence criterion for compatible stable assemblies. The precise local estimates, atlas definition, and compactness argument are given in Sections S2 and S4.

4 Spectral geometry as a universality quotient

The spectral map is broad but not injective. Three independent mechanisms make the loss of information explicit: directed current at fixed symmetric metric geometry, hidden dynamical memory at fixed static spectral data, and global marked monodromy at fixed local packets.

4.1 Information lost under spectralization

For a connected graph with fixed masses m and conductances c , the complete stationary current fibre is

$$\mathcal{J}(G, c) = \{j \in \text{Ker } B : |j_e| < c_e \text{ for every } e\}, \quad \dim \mathcal{J}(G, c) = b_1(G). \quad (29)$$

Every point of this fibre has the same metric spectral triple, while path probabilities, time-reversal asymmetry, and entropy production can vary. The lost directionality is recovered by the modular affinity

$$a_e = \operatorname{arctanh} \frac{j_e}{c_e} = \frac{1}{2} \log \frac{q_e}{q_{\bar{e}}}, \quad (30)$$

whose cohomology class parametrizes the current fibre.

Hidden positive memory gives a second fibre. Eliminating auxiliary degrees of freedom produces a Stieltjes self-energy

$$\Sigma_\mu(z) = \int_{(0,\infty)} \frac{1}{\lambda - z} d\mu(\lambda) = B_\mu(C_\mu - z)^{-1} B_\mu^*. \quad (31)$$

The full operator-valued measure μ classifies the minimal positive dilation, whereas a static spectral head may see only selected boundary values or moments. Finite rational Hermitian indefinite memory has the corresponding minimal Pontryagin/descriptor classification.

The third fibre is global. Locally marked-isomorphic spectral packets glued over a connected incidence graph are classified by

$$[\rho] \in \operatorname{Hom}(\pi_1(\Gamma), G_S)/G_S. \quad (32)$$

Trivial monodromy gives a global marked trivialization; nontrivial monodromy gives a flat twisted spectral geometry.

4.2 Universality fibres and coarse graining

These fibres have canonical effective images. Current is pushed through the graph chain map; high-energy positive memory is replaced by an exact zero-energy contact correction plus the retained low-energy measure; and monodromy descends precisely when the collapsed cycles lie in its kernel. The maps compose under successive coarse grainings.

Theorem 4.1 (Universality fibres and coarse graining). *For the classes above:*

- (i) *the stationary realization fibre of a metric graph triple is $\mathcal{J}(G, c)$, and the modular class recovers the complete directed generator;*
- (ii) *positive hidden memory is classified by its minimal operator-valued Stieltjes measure, while finite rational Hermitian indefinite memory is classified by a minimal Pontryagin/descriptor realization;*
- (iii) *locally identical marked packets are globally classified by the monodromy class (32);*
- (iv) *current, memory, and monodromy admit canonical composable coarse-graining maps, with the corresponding entropy/rank/index monotonicity statements;*
- (v) *after auxiliary markings, calibration, and joint Gram data are retained, the fully reduced finite inverse fibre is a stratified stabilizer quotient, and under the explicit separation, tightness, domain, weight, and transport hypotheses of the supplement the completed fibre is the inverse limit of those finite strata.*

The detailed current geometry, operator-measure realization, monodromy classification, coarse-graining maps, rational indefinite memory, and completed inverse-fibre theorems are given in [Sections S3](#), [S5](#) and [S6](#).

5 Interpretation and discussion

The results separate predictive reconstruction from operator-theoretic spectralization in a useful way. Renewal Geometry first reconstructs the Grand Tensor $\mathbb{T}_{\text{Grand},X}$, including the future-minimal predictive carrier, represented history algebra, and positive moment hierarchy. Spectralization then equips that reconstructed algebra with a compatible first-order differential datum δ_X . Once $(\mathcal{A}, \tau, \delta)$ is given, the finite Hodge–Dirac theorem is an ordinary statement about C^* -algebras, Hilbert bimodules, and derivations. The predictive reconstruction is therefore logically prior to the spectral triple without being an extra hypothesis of the abstract operator-theoretic result.

5.1 Spectral geometry as an effective description of predictive dynamics

The first consequence is a change in the logical status of the spectral triple. In Connes’ formulation, the algebra, its Hilbert-space representation, and the Dirac operator constitute the spectral data from which geometry is extracted. In the present construction these objects can instead occur downstream of the Renewal Grand Tensor:

$$\text{predictive dynamics} \longrightarrow \mathbb{T}_{\text{Grand},X} \longmapsto (\mathcal{A}_X^{\text{hist}}, \tau_X) \xrightarrow{\delta_X} (\mathcal{A}_X, \mathcal{H}_X, D_X, J_X, \gamma_X). \quad (33)$$

A spectral triple therefore need not be interpreted as the most microscopic object in the description. Within the framework considered here, it is the spectral image of a specific algebraic and differential sector of a richer predictive object.

This does not mean that an arbitrary Dirac operator is determined by the algebra. The finite Hodge construction shows precisely where additional information enters. The algebra and faithful trace do not fix an absolute metric scale, and the commutator derivation determines the full compatible Dirac operator only modulo the appropriate self-adjoint commutant. Likewise, finite marked surjectivity is a statement about an enlarged reconstruction category in which the missing calibration coordinates are retained; it is not a claim that an arbitrary target Dirac operator has been derived from the austere Hodge datum alone. At the stricter norm-monoidal level the reconstruction becomes genuinely selective, with spectral quasidiagonality providing the exact criterion.

The independent metric construction is important for the same reason. On the periodic A_3 family, the Riemannian metric reconstructed from jump second moments and the Connes distance reconstructed from the Hodge–Dirac operator are obtained by different routes and converge to the same continuum geometry. The agreement therefore tests the spectralization rather than defining the target metric by construction. For general closed Riemannian manifolds the scalar metric approximation remains unconditional, while the spin norm-resolvent statement requires the compatible stable-discretization hypotheses stated above.

A common pregeometric origin of spacetime and spectral geometry. The interpretation is strengthened by the companion spacetime construction [8]. There, the same Renewal Grand Tensor $\mathbb{T}_{\text{Grand},X}$ supports a reconstruction of Lorentzian spacetime without requiring a spectral triple as an intermediate primitive. Together, the two papers therefore place the Grand Tensor as a common pregeometric object upstream of two distinct downstream descriptions, as summarized in (2).

This matters conceptually. If spectral geometry were the only downstream output, one could interpret the predictive construction merely as an alternative procedure for generating spectral triples. The existence of an independent spacetime branch instead suggests that Lorentzian geometry

and Connes-type spectral geometry are different projections of a common pregeometric structure rather than successive fundamental layers.

The claim is structural rather than ontological. The results do not establish that Renewal Geometry is the unique microscopic theory underlying all noncommutative geometries or all physical spacetimes. They show that, within the framework studied here, neither spectral geometry nor spacetime geometry needs to be placed at the primitive entrance. Both may arise from a predictive structure defined before either metric geometry or a spectral triple is introduced.

Information loss and nonfaithfulness. The inverse results sharpen this effective interpretation. Spectralization is generically many-to-one:

$$\mathfrak{S}(\mathcal{R}_1) = \mathfrak{S}(\mathcal{R}_2) \quad \text{need not imply} \quad \mathcal{R}_1 \simeq \mathcal{R}_2, \quad (34)$$

where $\mathcal{R}_i$ denote predictive or dynamical realizations and $\mathfrak{S}$ their spectral image. A spectral triple is therefore not, in general, a complete invariant of the underlying dynamics.

The stationary-current fibre provides the simplest example. Fixing the masses and symmetric conductances fixes the metric Hodge–Dirac triple, while a divergence-free stationary current remains undetermined. Moving within this fibre changes path probabilities, time-reversal asymmetry, and entropy production without changing the spectral metric. The spectral geometry retains the symmetric metric sector while forgetting part of the directed dynamics.

The memory fibre exhibits a different form of information loss. Eliminating auxiliary degrees of freedom produces a frequency-dependent self-energy whose minimal realization is encoded by an operator-valued measure. A static spectral description may retain only selected boundary values, moments, or an effective operator, so distinct dynamical tails can have the same static spectral image. What is lost here is temporal organization rather than merely a finite parameter.

Global marked monodromy provides a third mechanism. Locally equivalent spectral packets can differ in how their marked realizations are assembled globally. Consequently spectralization can forget directional, dynamical, and global information for mathematically different reasons.

These examples motivate the description of spectral geometry as an *information-reduced effective geometry*. This should not be interpreted as a claim that noncommutative geometry is mathematically incomplete. A spectral triple may completely encode the geometric observables for which it is designed. The point is that it need not completely encode the predictive dynamics from which it arose. A single spectral geometry can therefore represent an entire family of dynamically inequivalent microscopic realizations.

The analogy with effective descriptions in statistical physics is useful. A set of macroscopic variables may completely characterize an effective state while failing to determine a unique microscopic realization. Likewise, a spectral triple may fully characterize an effective metric and differential sector while identifying systems that remain distinct at the predictive level.

Why the quotient has a universality interpretation. The inverse fibre

$$\mathfrak{S}^{-1}(S) \quad (35)$$

is therefore more than a technical obstruction to inversion. It describes the microscopic information quotiented out in the passage to a fixed spectral packet.

Moreover, the discarded variables are themselves organized under effective reduction. Directed currents are pushed through the induced graph chain map; high-energy memory can be absorbed into a zero-energy contact correction while the retained low-energy measure survives; and marked

monodromy descends exactly when the cycles removed by the quotient carry trivial holonomy. These maps compose under successive coarse grainings.

This is the sense in which the spectral packet acts as a universality-level observable. Different microscopic systems can share the same spectral geometry because distinctions that remain meaningful at the predictive level no longer affect the operator-geometric structure retained at that level of description.

The coarse-graining language should nevertheless be interpreted precisely. The results do not prove that spectralization is universally generated by a renormalization-group flow, nor that arbitrary predictive dynamics converges toward a spectral fixed point. What is established is a controlled many-to-one map, explicit inverse fibres, and composable effective maps on their principal coordinates. The universality interpretation is therefore structural rather than a claim of a universal dynamical attractor.

5.2 Inverse geometry, scope, and outlook

Once the lost information has been identified, a natural question is whether it should be retained when it is physically relevant. The present examples suggest several possible enrichments. A metric spectral triple can be supplemented by a modular or current class to retain irreversible directed information; an operator-valued self-energy can retain frequency-dependent memory; and a marked holonomy class can retain global assembly data. Schematically,

$$(\mathcal{A}, \mathcal{H}, D, J, \gamma) \longrightarrow (\mathcal{A}, \mathcal{H}, D, J, \gamma; \text{current, memory, monodromy}, \dots). \quad (36)$$

The ordinary spectral packet is recovered by forgetting the additional coordinates.

This perspective changes the natural inverse problem. Rather than expecting one microscopic dynamics to correspond to one spectral geometry, one should generally expect a moduli space or stratified fibre of realizations. The finite stabilizer quotient and its completed inverse-limit version provide concrete examples of such an inverse geometry. The relevant microscopic question is therefore not necessarily

“Which dynamics is represented by this spectral triple?”

but rather

“Which class of predictive dynamics has this spectral triple as its common effective geometry?”

The companion spacetime construction makes this question still richer. Two predictive systems that have the same spectral image need not necessarily have the same image under another downstream geometric reconstruction. Conversely, systems could agree on a spacetime sector while differing in spectral or dynamical refinements. The joint fibres of the spectral and Lorentzian projections of Renewal Geometry therefore provide a natural object for future study.

Mathematical scope. Several limitations of the present results should remain explicit. The strict spectral image would benefit from an intrinsic characterization independent of a chosen spectral projection sequence. For spin geometry, the local Wilson construction provides the required analytic estimates, but an unconditional global realization theorem would require verification of the compatibility and stability hypotheses for every closed spin manifold. Noncompact and genuinely semifinite geometries require further control of domains, localization, and weights. The positive

Stieltjes memory classification has a complete finite rational indefinite analogue, but the unrestricted nonrational infinite-dimensional indefinite problem is not treated here.

Nor does the finite compact-quantum-metric-space result by itself imply quantum Gromov–Hausdorff, propinquity, or spectral-propinquity convergence in the continuum limit [5, 6, 7]. Such statements require uniform control of the full state-space Lip-norms and, for spectral propinquity, compatible control of the spectral data beyond the pure-state distance convergence proved for the A_3 approximants.

Open questions and broader interpretation. The effective-geometric viewpoint raises several further questions. Which parts of the inverse fibre can be reconstructed from natural enrichments of a spectral triple? Which fibre coordinates remain stable under broader classes of coarse graining? Can additional dynamical principles select canonical representatives inside a spectral fibre? And how much of the many-to-one structure persists in continuum classes substantially larger than the finite and controlled compact-resolvent settings considered here?

The coexistence of the spectral and spacetime branches suggests a second family of questions. One may ask when the two downstream reconstructions determine one another, when one retains information absent from the other, and which microscopic coordinates are invisible to both. More generally, one can ask whether other geometric or physical structures arise as additional quotients of the same predictive object.

Within the scope established in this paper, the hierarchy is nevertheless clear. Predictive dynamics carries temporal, directional, and global information before a metric or spectral triple is introduced. Renewal Geometry organizes that information in the pregeometric Grand Tensor $\mathbb{T}_{\text{Grand},X}$. Lorentzian spacetime can emerge from one downstream sector, while on the spectral branch $\mathbb{T}_{\text{Grand},X}$ canonically supplies the predictive history algebra and positive moment structure, which become Connes-type spectral geometry after a compatible first-order variation is specified. The latter retains the metric and differential structures relevant to its level while identifying classes of underlying dynamics that remain distinct microscopically.

Thus, within this framework, Connes-type spectral geometry is naturally interpreted as a stable effective geometry of a richer predictive dynamics: powerful and potentially complete for its geometric sector, but not in general a complete description of the dynamics from which that geometry emerges.

References

- [1] A. Connes, *Noncommutative Geometry*, Academic Press, San Diego, 1994.
- [2] A. Rennie and J. C. Várilly, Reconstruction of manifolds in noncommutative geometry, arXiv:math/0610418, revised 2008.
- [3] A. Connes, On the spectral characterization of manifolds, *Journal of Noncommutative Geometry* **7** (2013), no. 1, 1–82.
- [4] M. A. Rieffel, Metrics on state spaces, *Documenta Mathematica* **4** (1999), 559–600.
- [5] M. A. Rieffel, Gromov–Hausdorff distance for quantum metric spaces, *Memoirs of the American Mathematical Society* **168** (2004), no. 796, 1–65. doi:10.1090/memo/0796.
- [6] F. Latrémolière, The Gromov–Hausdorff propinquity for metric spectral triples, *Advances in Mathematics* **404** (2022), 108393. doi:10.1016/j.aim.2022.108393.

- [7] C. Farsi, F. Latrémolière, and J. Packer, Convergence of inductive sequences of spectral triples for the spectral propinquity, *Advances in Mathematics* **437** (2024), 109442. doi:10.1016/j.aim.2023.109442.
- [8] A. Pélissier, Renewal Geometry and the emergence of Lorentzian spacetime, HAL preprint hal-05744772, 2026. <https://hal.science/hal-05744772>
- [9] T. Krajewski, Classification of finite spectral triples, *Journal of Geometry and Physics* **28** (1998), 1–30. doi:10.1016/S0393-0440(97)00068-5.
- [10] E. Christensen and C. Ivan, Spectral triples for AF C^* -algebras and metrics on the Cantor set, *Journal of Operator Theory* **56** (2006), no. 1, 17–46.
- [11] J. Pearson and J. Bellissard, Noncommutative Riemannian geometry and diffusion on ultrametric Cantor sets, *Journal of Noncommutative Geometry* **3** (2009), no. 3, 447–480.
- [12] E. Christensen, C. Ivan, and E. Schrohe, Spectral triples and the geometry of fractals, *Journal of Noncommutative Geometry* **6** (2012), no. 2, 249–274. doi:10.4171/JNCG/91.
- [13] J. Kellendonk and J. Savinien, Spectral triples and characterization of aperiodic order, *Proceedings of the London Mathematical Society* **104** (2012), no. 1, 123–157.
- [14] A. L. T. Paterson, Contractive spectral triples for crossed products, *Mathematica Scandinavica* **114** (2014), no. 2, 275–298. doi:10.7146/math.scand.a-17112.
- [15] A. Julien and I. F. Putnam, Spectral triples for subshifts, *Journal of Functional Analysis* **270** (2016), no. 3, 1031–1063. doi:10.1016/j.jfa.2015.12.002.
- [16] V. Aiello, D. Guido, and T. Isola, Spectral triples on irreversible C^* -dynamical systems, *International Journal of Mathematics* **33** (2022), no. 1, 2250005. doi:10.1142/S0129167X22500057.
- [17] S. Majid, Dirac operator associated to a quantum metric, *Journal of Noncommutative Geometry* **19** (2025), no. 2, 573–600.
- [18] F. Cipriani and J.-L. Sauvageot, Derivations as square roots of Dirichlet forms, *Journal of Functional Analysis* **201** (2003), no. 1, 78–120.
- [19] M. Hinz and A. Teplyaev, Dirac and magnetic Schrödinger operators on fractals, *Journal of Functional Analysis* **265** (2013), no. 11, 2830–2854. doi:10.1016/j.jfa.2013.07.021.
- [20] M. Hinz, D. J. Kelleher, and A. Teplyaev, Metric and spectral triples for Dirichlet and resistance forms, *Journal of Noncommutative Geometry* **9** (2015), no. 2, 359–390. doi:10.4171/JNCG/195.
- [21] F. Cipriani, Noncommutative potential theory: A survey, *Journal of Geometry and Physics* **105** (2016), 25–59.
- [22] C. Arhancet, Spectral triples, Coulhon–Varopoulos dimension and heat kernel estimates, *Advances in Mathematics* **451** (2024), 109794.
- [23] G. Chiribella, G. M. D’Ariano, and P. Perinotti, Theoretical framework for quantum networks, *Physical Review A* **80** (2009), 022339.

- [24] F. A. Pollock, C. Rodríguez-Rosario, T. Frauenheim, M. Paternostro, and K. Modi, Non-Markovian quantum processes: Complete framework and efficient characterization, *Physical Review A* **97** (2018), 012127. doi:10.1103/PhysRevA.97.012127.
- [25] F. A. Pollock, C. Rodríguez-Rosario, T. Frauenheim, M. Paternostro, and K. Modi, Operational Markov condition for quantum processes, *Physical Review Letters* **120** (2018), 040405.
- [26] H. Jaeger, Observable operator models for discrete stochastic time series, *Neural Computation* **12** (2000), no. 6, 1371–1398. doi:10.1162/089976600300015411.
- [27] M. L. Littman, R. S. Sutton, and S. Singh, Predictive representations of state, in *Advances in Neural Information Processing Systems 14*, MIT Press, 2002, pp. 1555–1561.
- [28] S. Singh, M. R. James, and M. R. Rudary, Predictive state representations: A new theory for modeling dynamical systems, in *Proceedings of the 20th Conference on Uncertainty in Artificial Intelligence*, 2004, pp. 512–519.
- [29] J. P. Crutchfield and K. Young, Inferring statistical complexity, *Physical Review Letters* **63** (1989), 105–108. doi:10.1103/PhysRevLett.63.105.
- [30] C. R. Shalizi and J. P. Crutchfield, Computational mechanics: Pattern and prediction, structure and simplicity, *Journal of Statistical Physics* **104** (2001), 817–879.
- [31] N. F. Travers and J. P. Crutchfield, Equivalence of history and generator ϵ -machines, *Symmetry* **17** (2025), no. 1, 78. doi:10.3390/sym17010078.
- [32] S. E. Marzen and J. P. Crutchfield, Structure and randomness of continuous-time, discrete-event processes, *Journal of Statistical Physics* **169** (2017), 303–315.
- [33] M. A. Rieffel, Metrics on states from actions of compact groups, *Documenta Mathematica* **3** (1998), 215–230. doi:10.4171/DM/41.
- [34] H. B. Lawson and M.-L. Michelsohn, *Spin Geometry*, Princeton University Press, Princeton, 1989.
- [35] H. B. Nielsen and M. Ninomiya, Absence of neutrinos on a lattice. I. Proof by homotopy theory, *Nuclear Physics B* **185** (1981), 20–40.
- [36] H. B. Nielsen and M. Ninomiya, Absence of neutrinos on a lattice. II. Intuitive topological proof, *Nuclear Physics B* **193** (1981), 173–194.
- [37] S. Nakamura, Remarks on discrete Dirac operators and their continuum limits, *Journal of Spectral Theory* **14** (2024), no. 1, 255–269. doi:10.4171/JST/478.

Mathematical Supplement

The main text reconstructs the finite Renewal Grand Tensor from operational histories before introducing spectral data. Its algebraic sector canonically supplies the future-minimal predictive carrier, the represented history algebra, and its positive moment structure, while a compatible first-order variation is supplied separately. The mathematical supplement begins at this algebraic interface, since the spectralization theorems depend only on the resulting finite-dimensional unital C^* -algebra, faithful tracial state, and compatible derivation into a Hilbert bimodule.

The associated Hodge–Dirac construction yields a real even spectral triple, whose metric scale is not fixed by the algebra and trace alone, and whose Dirac operator is determined by its commutator derivation only modulo a compatible self-adjoint commutant. After adjoining finite calibrated markings, every finite spectral packet is reconstructible; every separable compact-resolvent packet arises as a strong cofinal limit of such finite reconstructions, while norm-monoidal realizability is equivalent to spectral quasidiagonality. For stationary Markov graphs, the Hodge–Dirac square is the symmetric generator and its commutator seminorm is the corresponding local quadratic form; on periodic A_3 approximants, the resulting Connes distance converges uniformly to the Riemannian metric reconstructed independently from the jump second moments. The same framework also makes the nonfaithfulness of spectralization explicit through stationary currents, dynamical memory, and global marked monodromy, together with their natural coarse-graining maps. The sections below give the precise definitions, hypotheses, constructions, and proofs underlying these results.

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S1 Finite spectralization and essential image

S1.1 Algebraic input and operational motivation

The finite construction requires only a finite-dimensional unital C^* -algebra $\mathcal{A}$, a faithful tracial state τ , and the differential datum specified below. No operational reconstruction theorem is assumed. The Hilbert space of zero-forms is the vector space $\mathcal{A}$ with inner product

$$\mathcal{H}_0 = L^2(\mathcal{A}, \tau), \quad \langle a, b \rangle_0 = \tau(a^*b). \quad (37)$$

Faithfulness makes this inner product nondegenerate. Left and right multiplication are bounded commuting actions, with adjoints $L_a^* = L_{a^*}$ and $R_a^* = R_{a^*}$. The right action is a representation of the opposite algebra: $R_a R_b = R_{ba}$.

There is a distinguished possible choice of trace, but the results do not require it. If $\mathcal{A} \cong \bigoplus_\lambda M_{n_\lambda}(\mathbb{C})$, its normalized regular trace is

$$\hat{\tau}_{\text{reg}}(a) = \frac{\text{Tr}_{\mathbb{C}}(L_a)}{\dim_{\mathbb{C}} \mathcal{A}} = \frac{\sum_\lambda n_\lambda \text{Tr}(a_\lambda)}{\sum_\lambda n_\lambda^2}. \quad (38)$$

This is a faithful tracial state invariant under $*$ -isomorphisms. Thus both $\mathcal{A}$ and $\hat{\tau}_{\text{reg}}$ have ordinary finite-dimensional algebraic meanings independent of the motivating history construction.

A Hilbert $\mathcal{A}$ -bimodule means a Hilbert space $\mathcal{K}$ with commuting unital $*$ -representations of $\mathcal{A}$ and $\mathcal{A}^{\text{op}}$. We denote these by $\lambda_{\mathcal{K}}(a)$ and $\rho_{\mathcal{K}}(b)$, so that $a\xi b = \lambda_{\mathcal{K}}(a)\rho_{\mathcal{K}}(b)\xi$ and $\rho_{\mathcal{K}}(a)\rho_{\mathcal{K}}(b) = \rho_{\mathcal{K}}(ba)$.

Definition S1.1 (Finite real differential datum). A finite real differential datum on $(\mathcal{A}, \tau)$ consists of a Hilbert $\mathcal{A}$ -bimodule $\mathcal{K}$, an antiunitary involution $C_{\mathcal{K}}$ with

$$C_{\mathcal{K}}(a\xi b) = b^* C_{\mathcal{K}}(\xi) a^*,$$

and a complex-linear map $\delta : \mathcal{A} \rightarrow \mathcal{K}$ satisfying

$$\delta(\mathbf{1}) = 0, \quad \delta(ab) = a\delta(b) + \delta(a)b, \quad C_{\mathcal{K}}\delta(a) = \delta(a^*). \quad (39)$$

Only the bimodule generated by $\delta(\mathcal{A})$ is retained. The term *predictive differential datum* will occasionally be used when this same algebraic datum is obtained from the operational setting. No result in this section depends on how the datum was produced.

For completeness, the one-form construction can be performed entirely within this datum. The universal first-order calculus is

$$\Omega_u^1(\mathcal{A}) = \text{Ker}(m : \mathcal{A} \otimes \mathcal{A} \rightarrow \mathcal{A}), \quad d_u a = \mathbf{1} \otimes a - a \otimes \mathbf{1}, \quad (40)$$

with bimodule action $c(a \otimes b)d = ca \otimes bd$ and conjugate-linear reversal

$$\left(\sum_i a_i \otimes b_i \right)^{\natural} = - \sum_i b_i^* \otimes a_i^*. \quad (41)$$

Direct expansion gives $(d_u a)^\natural = d_u(a^*)$ and $d_u(ab) = ad_u b + (d_u a)b$. The map

$$T_\delta \left(\sum_i a_i \otimes b_i \right) = \sum_i a_i \delta(b_i), \quad \sum_i a_i b_i = 0,$$

is the unique bimodule map with $T_\delta d_u = \delta$. Indeed, the Leibniz rule makes it right linear because the additional term is $(\sum_i a_i b_i) \delta(c) = 0$; left linearity is immediate. Star compatibility gives $T_\delta(\omega^\natural) = C_\mathcal{K} T_\delta(\omega)$. Consequently

$$Q_\delta(\omega, \eta) = \langle T_\delta \omega, T_\delta \eta \rangle_\mathcal{K} \quad (42)$$

is positive, and its null space is exactly $\text{Ker } T_\delta$. The resulting one-form Hilbert space is

$$\mathcal{H}_1(\delta) = \Omega_u^1(\mathcal{A}) / \text{Ker } T_\delta \cong \text{Ran } T_\delta = \text{span}\{a\delta(b)c : a, b, c \in \mathcal{A}\}. \quad (43)$$

It is finite dimensional even if the initially supplied $\mathcal{K}$ is not. The displayed identification is isometric, so this quotient removes exactly the unused bimodule directions. This is the meaning of *source minimality* here. Write $\partial a = [d_u a]$ for the induced derivation.

Predictive histories provide a motivation and a class of examples, rather than extra axioms for this construction. Section 2.1 reconstructs self-containedly the finite Renewal Grand Tensor $\mathbb{T}_{\text{Grand}, X}$ from operational histories. In particular, $\mathbb{T}_{\text{Grand}, X}$ canonically contains the separated history algebra $\mathcal{A}_X^{\text{hist}}$ and its positive word-moment hierarchy; the normalized regular trace is an intrinsic faithful tracial state on that finite algebra. A compatible first-order variation δ_X is then supplied as the additional differential datum to be spectralized, as described in Section 2.1. Once $(\mathcal{A}, \tau, \delta)$ is given, the finite results below use only the algebraic data of Definition S1.1.

S1.2 The associated spectral triple

Set $\mathcal{H}_0 = L^2(\mathcal{A}, \tau)$ and let L_a and R_a denote left and right multiplication on $\mathcal{H}_0$. On

$$\mathcal{H} = \mathcal{H}_0 \oplus \mathcal{H}_1(\delta) \quad (44)$$

define

$$\pi(a) = L_a \oplus \lambda(a), \quad D = \begin{pmatrix} 0 & \partial^* \\ \partial & 0 \end{pmatrix}, \quad \gamma = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}. \quad (45)$$

On $\mathcal{H}_0$ put $J_0 a = a^*$, and on $\mathcal{H}_1(\delta)$ let $J_1[\omega] = [\omega^\natural]$; set $J = J_0 \oplus J_1$.

Theorem S1.2 (Finite spectralization). *The quintuple $(\mathcal{A}, \mathcal{H}, D, J, \gamma)$ is a real even finite spectral triple. More precisely, π is faithful, D is self-adjoint, J is antiunitary,*

$$J^2 = I, \quad JD = DJ, \quad J\gamma = \gamma J,$$

and the opposite representation is

$$\pi^o(b) = J\pi(b^*)J^{-1} = R_b \oplus \rho(b).$$

For all $a, b \in \mathcal{A}$,

$$[\pi(a), \pi^o(b)] = 0, \quad [[D, \pi(a)], \pi^o(b)] = 0. \quad (46)$$

If $B_a : \mathcal{H}_0 \rightarrow \mathcal{H}_1(\delta)$ is defined by $B_a b = (\partial a)b$, then

$$[D, \pi(a)] = \begin{pmatrix} 0 & -B_{a^*}^* \\ B_a & 0 \end{pmatrix}. \quad (47)$$

In particular, for $a = a^*$,

$$\|[D, \pi(a)]\| = \|B_a\|, \quad \text{Ker}(a \mapsto \|[D, \pi(a)]\|) = \text{Ker } \partial \cap \mathcal{A}_{\text{sa}}. \quad (48)$$

Proof. Faithfulness follows from the left regular summand. Since all spaces are finite dimensional, D is self-adjoint with compact resolvent. The Leibniz rule gives

$$(\partial L_a - \lambda(a)\partial)b = \partial(ab) - a\partial b = (\partial a)b = B_a b,$$

and taking adjoints yields (47). The reversal identities make J_1 antiunitary and imply $J_1\partial = \partial J_0$, hence $JD = DJ$. Moreover $J_0 L_b^* J_0 = R_b$ and $J_1 \lambda(b^*) J_1 = \rho(b)$, so the opposite representation has the asserted form. Left and right actions commute. Finally, the lower-left block of the double commutator is $B_a R_b - \rho(b) B_a$, which vanishes because B_a is right $\mathcal{A}$ -linear. The upper-right block is its adjoint. For self-adjoint a , the two off-diagonal blocks are adjoints, proving (48). $\square$

Remark S1.3. The real structure in Theorem S1.2 is the algebraic reversal associated with the history bimodule. Additional physical interpretations of J require extra markings and are not used in the mathematical results below.

Proposition S1.4 (No algebra-and-trace determination of the Dirac scale). *Let $\delta \neq 0$ be a finite real differential datum and let $s > 0$. Replacing δ by $s\delta$ leaves $(\mathcal{A}, \tau)$ unchanged and, under the canonical identification of the one-form carriers, sends*

$$D \mapsto sD, \quad L_D(a) := \|[D, \pi(a)]\| \mapsto sL_D(a). \quad (49)$$

Whenever the Connes distance is finite, it scales as $d_D \mapsto s^{-1}d_D$. Hence the algebra and canonical trace do not determine an absolute spectral metric scale.

Proof. The inner product induced by $s\delta$ is $s^2 Q_\delta$, so the canonical unitary from the scaled one-form carrier to the original one sends the derivation vector $[d_u a]$ to $s[d_u a]$. Both off-diagonal blocks of D therefore acquire the factor s . The commutator and dual-distance scalings follow immediately. $\square$

The spectral seminorm also gives a compact quantum metric space, provided that zero differential means constant observable. This is the finite-dimensional form of the state-space criterion underlying Rieffel's theory [33, 4].

Proposition S1.5 (Compact quantum metric-space structure). *Suppose $\text{Ker } \partial = \mathbb{C}\mathbf{1}$ and set $L_D(a) = \|[D, \pi(a)]\|$. On the real order-unit space $\mathcal{A}_{\text{sa}}$, L_D is a continuous Lip-norm. Explicitly, the distance*

$$\text{mk}_{L_D}(\varphi, \psi) = \sup\{|\varphi(a) - \psi(a)| : a = a^*, L_D(a) \leq 1\} \quad (50)$$

is finite on the state space $S(\mathcal{A})$ and induces its weak- topology. Moreover, the complex extension satisfies*

$$L_D(ab) \leq \|a\|L_D(b) + \|b\|L_D(a), \quad L_D(a^*) = L_D(a). \quad (51)$$

Conversely, if $\text{Ker } \partial \neq \mathbb{C}\mathbf{1}$, the distance in (50) is infinite for some pair of states, so it is not a compact quantum metric on the whole state space.

Proof. The commutator is a linear map between finite-dimensional normed spaces; hence L_D is continuous. The commutator product rule and the *-identity give (51). By Theorem S1.2, the kernel on $\mathcal{A}_{\text{sa}}$ is $\mathbb{R}\mathbf{1}$. On $E_\tau = \{a = a^* : \tau(a) = 0\}$ it follows that L_D is a norm, and finite-dimensional norm equivalence gives a constant C_δ such that

$$\|a - \tau(a)\mathbf{1}\| \leq C_\delta L_D(a), \quad a = a^*. \quad (52)$$

Thus $\{a \in E_\tau : L_D(a) \leq 1\}$ is norm compact. Scalar multiples of $\mathbf{1}$ do not affect differences of states, so the supremum in (50) can be taken over this compact set and is at most $2C_\delta$. Weak-*

convergence of states is uniform on a norm-compact set, since states have norm one; it therefore implies convergence in mk_{L_D} . Conversely, for each self-adjoint a ,

$$|\varphi(a) - \psi(a)| \leq L_D(a) \text{mk}_{L_D}(\varphi, \psi),$$

with the scalar case immediate. Hence metric convergence implies weak-* convergence and the metric separates states.

If the kernel is larger, star compatibility gives a non-scalar self-adjoint $b \in \text{Ker } \partial$. States separate self-adjoint elements modulo scalars, so some φ, ψ have $\varphi(b) \neq \psi(b)$. Every multiple tb has zero seminorm, making their distance infinite. $\square$

The quotient used here is $\mathcal{A}_{\text{sa}}/\mathbb{R}\mathbf{1}$; the metric itself is on $S(\mathcal{A})$, not on a quotient of states. Finite dimensionality is essential to the automatic compactness argument: for an infinite-dimensional limit, a scalar kernel alone does not replace total boundedness of the normalized Lipschitz ball.

Proposition S1.6 (Unit zero-mode of the austere Hodge construction). *For every differential datum of Definition S1.1, the Hodge operator of Theorem S1.2 satisfies*

$$D(\mathbf{1}, 0) = 0. \quad (53)$$

Consequently the bare Hodge functor $\mathfrak{H}_{\text{fin}}$ never produces an invertible finite Dirac operator. More generally, knowledge of the commutator derivation does not determine the component of D in the compatible commutant; that residual ambiguity is characterized in Theorem S1.7.

Proof. The universal derivation satisfies $\partial\mathbf{1} = 0$, hence the lower-left block of $D(\mathbf{1}, 0)$ vanishes; the upper-right block acts on the zero one-form component and vanishes as well. The second assertion is immediate from Theorem S1.7. $\square$

S1.3 The commutant fibre of a fixed derivation

Let $\mathbf{S} = (\mathcal{A}, \mathcal{H}, \pi, D, J, \gamma)$ be a finite marked spectral packet. Define

$$\mathfrak{c}_{\mathbf{S}} = \{C \in \pi(\mathcal{A})'_{\text{sa}} : C\gamma + \gamma C = 0, \quad JC = \varepsilon' C J\}, \quad (54)$$

with the evident omission of a condition when the corresponding mark is absent.

Theorem S1.7 (Exact Dirac commutant fibre). *Among self-adjoint operators with the declared grading and reality signs that implement the same commutator derivation $[D, \pi(\cdot)]$, the complete solution set is the affine space*

$$D + \mathfrak{c}_{\mathbf{S}}. \quad (55)$$

Let $P_{\mathfrak{c}_{\mathbf{S}}}$ denote Hilbert–Schmidt projection onto $\mathfrak{c}_{\mathbf{S}}$ and put

$$D_{\text{der}} = D - P_{\mathfrak{c}_{\mathbf{S}}} D. \quad (56)$$

Then D_{der} is the unique minimum-Hilbert–Schmidt-norm representative of the affine fibre. Moreover $q_{\mathbf{S}} = \dim_{\mathbb{R}} \mathfrak{c}_{\mathbf{S}}$ independent real linear scalar measurements are necessary and sufficient to distinguish every point of the fibre.

Proof. If D' has the same commutator derivation as D , then $C = D' - D$ commutes with $\pi(\mathcal{A})$. Subtracting the grading and reality equations for D' and D shows that $C \in \mathfrak{c}_{\mathbf{S}}$. Conversely every

$C \in \mathfrak{c}_S$ preserves the commutators and the declared signs. Thus the solution set is (55). Orthogonal projection gives

$$\|D + C\|_{\text{HS}}^2 = \|D_{\text{der}}\|_{\text{HS}}^2 + \|P_{\mathfrak{c}_S} D + C\|_{\text{HS}}^2,$$

which proves the minimum statement. Finally, a bank of scalar anchors restricts to a real linear map on $\mathfrak{c}_S$; injectivity requires at least q_S coordinates and is achieved by any basis of the dual space. $\square$

S1.4 Functoriality and essential image

Let $\mathbf{RenDiff}_{\text{fin}}$ have objects $(\mathcal{A}, \tau, \delta)$ as above and morphisms the trace-preserving $*$ -isomorphisms intertwining the differential bimodule data. The construction of Theorem S1.2 is functorial: the algebra isomorphism induces unitaries on $L^2(\mathcal{A}, \tau)$ and on the minimal one-form quotient, and these unitaries intertwine π, D, J, γ . We denote this austere Hodge functor by $\mathfrak{H}_{\text{fin}}$.

The marked reconstruction considered next is a different, deliberately enlarged map. It retains a represented carrier together with finitely many calibrated operator marks in addition to the differential datum; we denote it by $\mathfrak{S}_{\text{fin}}^{\text{mark}}$. The distinction is essential. The Hodge operator of Theorem S1.2 annihilates $(1, 0)$, so the bare functor $\mathfrak{H}_{\text{fin}}$ cannot be unitarily equivalent to an arbitrary invertible finite Dirac operator. The marked image statement below concerns the enlarged calibrated reconstruction, not surjectivity of the Hodge functor.

For the inverse problem we distinguish three categories. In the *finite marked category*, a finite target may retain finitely many calibration and marking data. In the *strong compact category*, finite-rank spectral compressions converge strongly on fixed vectors, resolvents, commutators, and matrix coefficients. In the *strict norm-monoidal category*, the finite compressions must additionally transport multiplication asymptotically in norm. A compact-resolvent packet is called *spectrally quasidiagonal* if there exist finite-rank projections $P_n \uparrow I$ with

$$P_n D = D P_n, \quad \|[P_n, \pi(a)]\| \rightarrow 0 \quad (57)$$

for every a in a dense spectral algebra, equivariantly for J and γ when these are marked.

Theorem S1.8 (Calibrated marked image and norm-compatible essential image). *The following statements concern the enlarged calibrated reconstruction in the first two lines and the norm-compatible approximation problem in the third. They do not assert that the austere Hodge functor $\mathfrak{H}_{\text{fin}}$ is surjective:*

$$\begin{aligned} \text{EssIm}(\mathfrak{S}_{\text{fin}}^{\text{mark}}) &= \mathbf{Spec}_{\text{fin}}, \\ \text{EssIm}(\mathfrak{S}_{\text{cpt}}^{\text{strong, mark}}) &= \mathbf{Spec}_{\text{cpt}}^{\text{sep}}, \\ \text{EssIm}(\mathfrak{S}_{\text{cpt}}^{\text{strict}}) &= \mathbf{Spec}_{\text{cpt}}^{\text{sqd}} \subsetneq \mathbf{Spec}_{\text{cpt}}^{\text{sep}}. \end{aligned} \quad (58)$$

Thus every finite spectral packet is reconstructible once the finite calibration marks defining the marked category are retained; every separable compact-resolvent packet is a strong cofinal limit of such finite marked reconstructions; and strict norm-monoidal realizability is equivalent to spectral quasidiagonality.

Proof. For a finite target $(\mathcal{B}, K, \pi_0, D_0, J_0, \gamma_0)$, choose finitely many self-adjoint generators of $\pi_0(\mathcal{B})$. Their exponentials, with angles chosen so that functional calculus recovers the generators, give a finite family of generating unitary matrices. Retain their full matrix-word coefficients on K , the declared J_0, γ_0 marks, and the calibrated unitary germ $U(t) = e^{-itD_0}$. At this point D_0 is deliberately part of the calibration data: the identity $i\dot{U}(0) = D_0$ is a reconstruction of the marked generator, not a derivation of D_0 from $\mathcal{B}$ or from a Hodge differential. If only the conjugation germ

is retained, its derivative determines the commutator action instead, and the missing compatible commutant coordinates are exactly those of Theorem S1.7. Evaluation of these finite-dimensional marked data recovers the target up to unitary isomorphism. The first equality in (58) should therefore be read as a calibrated reconstruction benchmark: it identifies which finite information closes the gap from the austere derivation image to an arbitrary marked target. The predictive entrance and its Grand-Tensor interface are reconstructed self-containedly in Section 2.1; the companion Renewal Geometry work [8] develops the same predictive construction in the broader Lorentzian-spacetime setting.

Let now $(\mathcal{A}, \mathcal{H}, \pi, D)$ be separable with compact resolvent. Let P_n be the increasing finite spectral projections of D . Then $P_n D = D P_n$ and $P_n \rightarrow I$ strongly. For every fixed a, b ,

$$P_n \pi(ab) P_n - P_n \pi(a) P_n \pi(b) P_n = P_n \pi(a) (I - P_n) \pi(b) P_n \quad (59)$$

tends strongly to zero. Here the finite-stage observables are the compressed matrices $P_n \pi(a) P_n$; the compression map from the original algebra is not asserted to be a $*$ -homomorphism at finite n . This is precisely why its product defect is retained. Applying the finite marked construction to these matrix data and $P_n D P_n$ yields strong convergence of fixed resolvents, commutators, and matrix coefficients, proving the second equality.

If (57) holds, (59) tends to zero in norm, so the spectral packet lies in the strict image. Conversely, norm multiplicativity applied to (a^*, a) gives

$$\|P_n \pi(a)^* (I - P_n) \pi(a) P_n\| \rightarrow 0,$$

which is exactly $\|(I - P_n) \pi(a) P_n\|^2 \rightarrow 0$. Applying the same argument to a^* controls the opposite off-diagonal block, hence $\|[P_n, \pi(a)]\| \rightarrow 0$. This proves equality with the spectrally quasidiagonal class.

Properness is seen from the Toeplitz representation on $\ell^2(\mathbb{N}_0)$ with number operator $N e_k = k e_k$ and unilateral shift $S e_k = e_{k+1}$. Any finite-rank projection commuting with N is diagonal in the number basis, and every nontrivial finite diagonal projection has $\|[P, S]\| = 1$. Hence this compact-resolvent spectral packet lies in the strong image but not in the strict one. $\square$

Remark S1.9. The first equality in (58) is intentionally not an “emergence from the algebra” statement. The full generator germ is allowed as a calibration mark, so finite surjectivity by itself is formal. The nontrivial rigidity content is instead the comparison with the austere Hodge image: Proposition S1.6 gives its unavoidable unit zero-mode, and Theorem S1.7 identifies exactly what a fixed commutator derivation fails to determine. The strict norm-monoidal line adds a separate, genuine approximation constraint.

S2 Metric realization and compact spin geometry

The essential-image theorem is a structural existence result. To test whether the spectralization also has geometric content, one needs a metric reconstructed by a route that does not use the Dirac operator. Stationary graph dynamics provides a particularly transparent setting because the symmetric jump covariance determines such an independent metric quantity.

S2.1 Stationary graphs and the Hodge–Dirac operator

Definition S2.1 (Stationary finite renewal graph). Let V be finite, let $m_x > 0$ with $\sum_x m_x = 1$, and let $k_{xy} \geq 0$ for $x \neq y$. Put

$$(\mathcal{L}f)(x) = \sum_{y \neq x} k_{xy}(f(y) - f(x)), \quad q_{xy} = m_x k_{xy}, \quad (60)$$

and assume stationarity

$$\sum_y (q_{xy} - q_{yx}) = 0 \quad (x \in V). \quad (61)$$

Define

$$c_{xy} = \frac{q_{xy} + q_{yx}}{2} = c_{yx}, \quad j_{xy} = \frac{q_{xy} - q_{yx}}{2} = -j_{yx}. \quad (62)$$

Assume the undirected graph on the pairs with $c_{xy} > 0$ is connected.

Set

$$\mathcal{H}_0 = \ell^2(V, m), \quad \mathcal{H}_1 = \ell^2(E^{\text{or}}, c), \quad (\partial f)(x, y) = f(y) - f(x), \quad (63)$$

where both orientations are retained and $\|\omega\|_1^2 = \sum_{x,y} c_{xy} |\omega(x, y)|^2$. Let

$$(\lambda(a)\omega)(x, y) = a(y)\omega(x, y), \quad (\rho(a)\omega)(x, y) = a(x)\omega(x, y). \quad (64)$$

Then $\partial(fg) = \lambda(f)\partial g + \rho(g)\partial f$. The associated Hodge operator is

$$D_{m,c} = \begin{pmatrix} 0 & \partial^* \\ \partial & 0 \end{pmatrix}. \quad (65)$$

Theorem S2.2 (Renewal Hodge square and exact Lipschitz seminorm). *Let $\mathcal{L}^\dagger$ be the adjoint of $\mathcal{L}$ in $\ell^2(V, m)$ and let $\mathcal{L}_s = (\mathcal{L} + \mathcal{L}^\dagger)/2$. Then*

$$D_{m,c}^2|_{\mathcal{H}_0} = \partial^* \partial = -2\mathcal{L}_s. \quad (66)$$

For every real $f \in C(V)$,

$$L_{m,c}(f)^2 = \|[D_{m,c}, f]\|^2 = \max_{x \in V} \frac{1}{m_x} \sum_y c_{xy} |f(y) - f(x)|^2. \quad (67)$$

The kernel of $L_{m,c}$ is the constants, so the Connes formula gives a finite metric on V . On the reversible branch, $\mathcal{L} = \mathcal{L}_s$ and $D_{m,c}^2|_{\mathcal{H}_0} = -2\mathcal{L}$.

Proof. The adjoint generator has rates q_{yx}/m_x , so $\mathcal{L}_s$ has rates c_{xy}/m_x . Therefore

$$-\langle f, \mathcal{L}_s f \rangle_0 = \frac{1}{2} \sum_{x,y} c_{xy} |f(y) - f(x)|^2 = \frac{1}{2} \|\partial f\|_1^2.$$

Polarization gives (66). The lower-left commutator block satisfies

$$(B_f g)(x, y) = (f(y) - f(x))g(x).$$

Hence

$$\|B_f g\|_1^2 = \sum_x |g(x)|^2 \sum_y c_{xy} |f(y) - f(x)|^2.$$

Its norm relative to $\sum_x m_x |g(x)|^2$ is the maximum in (67). Connectedness makes $\partial f = 0$ equivalent to constancy. $\square$

The connectedness assumption has a state-space consequence as well as a pointwise one. Proposition S1.5 implies that $(C(V)_{\text{sa}}, L_{m,c})$ is a compact quantum metric space. This finite-dimensional Lip-norm fact is standard and is not the new assertion of this section. The relevant structural identity is the exact equality (67) between the commutator seminorm and the local carré-du-champ of the symmetric Markov generator; the continuum result below is the uniform convergence of the resulting A_3 Connes metric to the independently reconstructed Riemannian metric. Its states are probability distributions on V , and the spectral metric on those distributions induces their usual weak-* topology. This conclusion does not require the Hodge operator itself to be invertible; it is the scalar kernel of the *commutator seminorm* that matters. The following large one-form kernel is therefore an issue for the operator continuum limit, not for the finite Lip-norm.

Proposition S2.3 (Extensive kernel of the one-step graph calculus). *Let the connected underlying graph have n vertices and e undirected edges, and retain both orientations in $\mathcal{H}_1$. Then*

$$\dim \text{Ker } D_{m,c} = 2e - n + 2. \quad (68)$$

On a periodic A_3 root graph, $e = 6n$, so the kernel has dimension $11n + 2$. Thus the one-step universal graph calculus cannot itself have a compact- resolvent cofinal limit retaining all one-form directions.

Proof. Connectedness gives $\dim \text{Ker } \partial = 1$ and $\text{rank } \partial = n - 1$. Since the directed one-form space has dimension $2e$, $\dim \text{Ker } \partial^* = 2e - n + 1$. Because $\text{Ker } D = (\text{Ker } \partial) \oplus (\text{Ker } \partial^*)$, the formula follows. $\square$

S2.2 The periodic A_3 model and convergence of the Connes metric

Let

$$W_0 = \left\{ x \in \mathbb{R}^4 : \sum_{i=1}^4 x_i = 0 \right\}, \quad \Lambda_{A_3} = W_0 \cap \mathbb{Z}^4, \quad (69)$$

and let $\Phi_{A_3} = \{e_i - e_j : i \neq j\}$ be the twelve oriented roots.

Lemma S2.4 (A_3 tight-frame identity). *On W_0 ,*

$$\sum_{\alpha \in \Phi_{A_3}} \alpha \alpha^\top = 8I_{W_0}. \quad (70)$$

Equivalently, $\sum_{i \neq j} (v_i - v_j)^2 = 8\|v\|^2$ for $v \in W_0$.

Proof. Expand the left side. Each v_i^2 occurs six times in the two diagonal sums, while $\sum_{i \neq j} v_i v_j = (\sum_i v_i)^2 - \sum_i v_i^2 = -\sum_i v_i^2$. The total is $8\sum_i v_i^2$. Polarization gives the operator identity. $\square$

Put

$$M = W_0 / \Lambda_{A_3}, \quad V_h = h\Lambda_{A_3} / \Lambda_{A_3}, \quad h = d^{-1}, \quad (71)$$

and assign each root edge the reversible rate and vertex mass

$$k_{\alpha,h} = \frac{1}{8h^2}, \quad m_h(x) = h^3, \quad c_{\alpha,h} = \frac{h}{8}. \quad (72)$$

Theorem S2.5 (Exact finite A_3 spectralization). *The graph spectral triple has*

$$L_h(f)^2 = \max_{x \in V_h} \frac{1}{8h^2} \sum_{\alpha \in \Phi_{A_3}} |f(x + h\alpha) - f(x)|^2, \quad (73)$$

$$D_h^2|_{C(V_h)} = -2\mathcal{L}_h, \quad (\mathcal{L}_h f)(x) = \frac{1}{8h^2} \sum_{\alpha \in \Phi_{A_3}} (f(x + h\alpha) - f(x)). \quad (74)$$

For every $F \in C^2(M)$,

$$|L_h(F|_{V_h}) - \|\nabla F\|_{L^\infty(M)}| \leq C_{A_3} h \|F\|_{C^2(M)}. \quad (75)$$

Moreover $\mathcal{L}_h F \rightarrow \frac{1}{2}\Delta F$ uniformly for $F \in C^4(M)$.

Proof. The first two identities follow from [Theorem S2.2](#) and [\(72\)](#). Taylor expansion gives

$$\frac{F(x + h\alpha) - F(x)}{h} = \alpha \cdot \nabla F(x) + r_{\alpha,h}(x), \quad |r_{\alpha,h}(x)| \leq Ch \|F\|_{C^2}.$$

Apply the reverse triangle inequality in $\ell^2(\Phi_{A_3})$ and [Lemma S2.4](#), then maximize over the mesh, to obtain [\(75\)](#). For the generator, the linear terms cancel by central symmetry. The quadratic term is $\frac{1}{16} \sum_{\alpha} (\alpha \otimes \alpha) : \nabla^2 F = \frac{1}{2} \Delta F$; the cubic terms cancel and the remainder is $O(h^2 \|F\|_{C^4})$. $\square$

Lemma S2.6 (Compactness of discrete unit-Lipschitz functions). *Let $h_n \downarrow 0$ and let $f_n : V_{h_n} \rightarrow \mathbb{R}$ satisfy $L_{h_n}(f_n) \leq 1$ and $f_n(x_{0,n}) = 0$ for $x_{0,n} \rightarrow x_0$. Then a subsequence of periodic piecewise-affine interpolants converges uniformly to $F \in W^{1,\infty}(M)$ with*

$$|\nabla F| \leq 1 \quad \text{almost everywhere.} \quad (76)$$

Proof. Equation [\(73\)](#) gives the root-edge bound $|f_n(x + h_n\alpha) - f_n(x)| \leq \sqrt{8}h_n$. A fixed shape-regular periodic simplicial refinement of the A_3 lattice therefore gives equi-Lipschitz, uniformly bounded interpolants. Arzelà–Ascoli gives uniform convergence along a subsequence.

The difference quotients $g_{n,\alpha} = (f_n(\cdot + h_n\alpha) - f_n)/h_n$ lie pointwise in the closed convex ball $\frac{1}{8} \sum_{\alpha} |z_{\alpha}|^2 \leq 1$. Passing weak-* in L^∞ and using discrete periodic summation by parts gives $g_{\alpha} = \alpha \cdot \nabla F$ distributionally. The tight-frame identity then yields $|\nabla F|^2 = \frac{1}{8} \sum_{\alpha} |\alpha \cdot \nabla F|^2 \leq 1$. $\square$

Theorem S2.7 (Renewal–spectral distance identity). *Let d_h be the pure-state Connes distance defined by [\(73\)](#), and let d_g be the flat geodesic distance of the metric independently reconstructed from the renewal second moment. Then*

$$\sup_{x,y \in V_h} |d_h(x,y) - d_g(x,y)| \longrightarrow 0. \quad (77)$$

Proof. For the lower bound, mollify the one-Lipschitz function $z \mapsto d_g(z,y)$ periodically to obtain smooth $F_{y,\varepsilon}$ with $\|F_{y,\varepsilon} - d_g(\cdot,y)\|_{\infty} \leq \varepsilon$ and $\|\nabla F_{y,\varepsilon}\|_{\infty} \leq 1$, uniformly in y . By [\(75\)](#), its sampled seminorm is at most $1 + o_h(1)$. The dual formula then gives $\liminf_h (d_h - d_g) \geq -2\varepsilon$ uniformly; let $\varepsilon \downarrow 0$.

If the upper bound failed, there would be $h_n \downarrow 0$, vertices x_n, y_n , and real f_n with $L_{h_n}(f_n) \leq 1$ and $f_n(x_n) - f_n(y_n) \geq d_g(x_n, y_n) + \varepsilon$. Normalize by $f_n(y_n) = 0$ and pass to subsequences. By [Lemma S2.6](#), the interpolants converge uniformly to a one-Lipschitz F . Thus $F(x) - F(y) \leq d_g(x,y)$, contradicting the limiting inequality. $\square$

Each finite A_3 approximant is consequently a compact quantum metric space. The distance convergence just proved concerns its pure states, that is, the vertices. It is not by itself a quantum Gromov–Hausdorff, propinquity, or spectral-propinquity convergence theorem: those stronger notions compare the full Lip-norms, state spaces, and, in the spectral setting, the corresponding operator data [5, 6, 7]. In particular, a quadratic graph commutator seminorm need not be the ordinary Lipschitz seminorm of its pure-state metric [4]. Nor does the finite-dimensional conclusion of Proposition S1.5 automatically survive an infinite-dimensional limit. A limiting quantum compact metric-space statement would require a uniform total-boundedness or equicontinuity estimate for normalized Lipschitz balls (the noncommutative analogue of the Arzelà–Ascoli input), together with control of the limiting seminorm. Lemma S2.6 supplies the required precompactness for the scalar functions used in the pure-state metric proof, but it is not asserted here to prove convergence of the full quantum state spaces. Keeping these assertions separate avoids conflating a pointwise metric limit with a noncommutative metric-space limit.

S2.3 Local spin discretization and norm-resolvent convergence under atlas stability

The periodic model makes the metric comparison explicit. We next formulate the local construction on an arbitrary closed Riemannian spin manifold $(M^d, g, \mathfrak{s})$ and separate what is proved chartwise from the additional global compatibility needed for norm-resolvent convergence. The local scalar ingredient is a positive decomposition of the inverse metric.

Lemma S2.8 (Positive local stencil). *Let*

$$\mathcal{R}_d^+ = \{e_i : 1 \leq i \leq d\} \cup \{e_i + e_j, e_i - e_j : 1 \leq i < j \leq d\}.$$

After a linear normalization at a point, every inverse metric sufficiently close to the identity admits a decomposition

$$g^{-1}(x) = \sum_{r \in \mathcal{R}_d^+} w_r(x) r r^\top \quad (78)$$

with all $w_r(x) > 0$. Hence a finite good cover of a closed Riemannian manifold may be chosen so that such a positive stencil exists on every chart.

Proof. Write $A = (A_{ij})$ for the normalized inverse metric and fix $\eta > 0$ such that $|A_{ij}| < \eta$ for $i \neq j$ and $A_{ii} > (d-1)\eta$. Set

$$w_i = A_{ii} - (d-1)\eta, \quad w_{ij}^\pm = \frac{1}{2}(\eta \pm A_{ij}).$$

All coefficients are positive and direct expansion gives

$$A = \sum_i w_i e_i e_i^\top + \sum_{i < j} [w_{ij}^+(e_i + e_j)(e_i + e_j)^\top + w_{ij}^-(e_i - e_j)(e_i - e_j)^\top].$$

Continuity and compactness then give a finite cover. □

The weights in (78) define reversible local jump processes whose second moments converge to g^{-1} and whose generators converge to $\frac{1}{2}\Delta_g$. To recover spin geometry one chooses the marked oriented orthonormal coframe and its spin lift. This is additional information: the scalar metric bracket

alone does not determine the spin structure. With $c^j = E_a^j \gamma_a$ and $\rho = \sqrt{\det g}$, the geometric Dirac operator admits the density-symmetric representation

$$\rho^{1/2} D_g \rho^{-1/2} = -\frac{i}{2} \sum_{j=1}^d (M_{c^j} \nabla_j + \nabla_j M_{c^j}), \quad (79)$$

which is well suited to self-adjoint finite-difference discretization [34].

A naive centered lattice Dirac operator generally has extra low-energy species; we therefore use a covariant Wilson term, as in the standard resolution of the fermion-doubling problem [35, 36]. Its role in rigorous continuum limits is discussed in [37]. For fixed positive Wilson parameter, this term is $O(h)$ on sampled smooth spinors but of size $O(h^{-1})$ at nonzero lattice corners. It removes the unwanted ultraviolet species without adding a term to the limiting equation on the smooth core. The distinction between these two estimates is essential: the Wilson term does not tend to zero in operator norm on the whole lattice space. A fixed two-dimensional normal Clifford factor supplies the stable real-even convention.

Passing from local coordinate estimates to a closed manifold requires more than a positive overlap Gram. The assembled finite operators must have a uniform first-order graph estimate, compatible reconstructions into the fixed spin bundle, and consistency on its smooth core. These requirements are stated precisely in Definition S4.7. They also specify which self-adjoint continuum realization is intended: the closure on smooth spinors, with domain H^1 , rather than independent boundary realizations on the coordinate charts.

Theorem S2.9 (Norm-resolvent convergence criterion under atlas stability). *Let $(M^d, g, \mathfrak{s})$ be a closed Riemannian spin manifold, and suppose that its finite covariant Wilson operators form a compatible stable spin discretization in the sense of Definition S4.7. Then, for every $z \in \mathbb{C} \setminus \mathbb{R}$,*

$$\left\| W_h(D_h - z)^{-1} W_h^* - (\hat{D}_{M, \mathfrak{s}} - z)^{-1} \right\| \longrightarrow 0, \quad \hat{D}_{M, \mathfrak{s}} = D_{M, \mathfrak{s}} \otimes \sigma_1. \quad (80)$$

The real and grading structures are retained when intertwined as specified there, and the global construction uses the fixed spin structure and orientation. Independently, every closed Riemannian manifold admits finite metric triples on refining meshes with

$$\sup_{x, y \in V_h} |d_h^{\text{met}}(x, y) - d_g(x, y)| \longrightarrow 0. \quad (81)$$

Thus the spin conclusion is a convergence theorem for compatible stable assemblies; the scalar metric approximation does not require those additional spin-assembly hypotheses.

Proof. Lemma S2.8 supplies reversible local generators whose scalar brackets approximate g^{-1} . On the spin side, the frozen Wilson square controls the full forward-difference symbol uniformly, including nonzero Brillouin corners. Appendix S4 gives the explicit symbol comparison and derives the chartwise Gårding bound by localization on fixed, sufficiently small coordinate patches.

In a compatible global assembly, smooth cutoffs with $\sum_\alpha \chi_\alpha^2 = 1$ introduce no order-one boundary term: $\sum_\alpha \chi_\alpha [\hat{D}, \chi_\alpha] = 0$ after the spin and density transitions are included. The analogous Wilson double-commutator error is $O(h)$ on common meshes. For independently meshed charts, the interpolation and assembled stability requirements are part of the explicit atlas hypotheses, not a consequence of positivity alone.

Self-adjointness bounds the discrete resolvents by $|\text{Im } z|^{-1}$. The global graph estimate and reconstruction bounds make their images collectively compact in L^2 . Testing the resolvent equation

against sampled smooth spinors identifies every limit with $(\widehat{D} - z)^{-1}$; the same argument at $\bar{z}$ gives adjoint convergence. These facts yield norm convergence by the compactness argument proved in Theorem S4.8.

For the metric conclusion, connect mesh points at distances of a mesoscopic scale $\rho_h \downarrow 0$, with $h/\rho_h \rightarrow 0$, using endpoint blocks weighted by the lengths of short Riemannian curves. The commutator seminorm is the ordinary graph Lipschitz seminorm. Approximating minimizing geodesics by mesh paths gives, uniformly,

$$d_g(x, y) \leq d_h^{\text{met}}(x, y) \leq d_g(x, y) + C \left(\rho_h + \frac{h}{\rho_h} \right).$$

This proves (81); the complete metric construction is in Theorem S4.9. $\square$

Remark S2.10. The theorem is Riemannian. It does not require, and does not assert, a Lorentzian or interacting quantum-field-theoretic completion. The role of the fixed normal Clifford factor is only to provide a stable real-even lattice realization. The local construction proves the necessary chartwise estimates. An unconditional assertion that a particular atlas construction realizes every closed spin manifold additionally requires verification of the global compatibility and stability properties in Definition S4.7. The convergence criterion does not assume that this follows merely from an overlap Gram quotient.

S3 Nonfaithfulness and inverse fibres

The preceding sections show that the spectralization map has a broad essential image. It is nevertheless far from injective. Everything in this section is formulated in standard operator-theoretic terms once the graph generator, the operator-valued self-energy, or the marked spectral packet is specified; the Renewal terminology records one motivating interpretation and is not an additional hypothesis. We describe three independent mechanisms of nonfaithfulness: directed current at fixed symmetric metric data, hidden dynamical memory, and global marked monodromy. Each admits an intrinsic coordinate, and the principal coordinates transform functorially under coarse graining.

S3.1 Directed currents at fixed metric geometry

Fix a connected undirected graph $G = (V, E)$, positive masses m_x , and positive conductances c_e . Choose one orientation of each undirected edge and let $B : \mathbb{R}^E \rightarrow \mathbb{R}^V$ be the incidence matrix. An oriented current is extended to the reverse orientation by $j_{\bar{e}} = -j_e$.

Definition S3.1 (Admissible cycle-current polytope). Define

$$\mathcal{J}(G, c) = \{j \in \text{Ker } B : |j_e| < c_e \text{ for every } e \in E\}. \quad (82)$$

Allowing non-strict inequalities gives the closed fibre in which some directed rates may vanish.

Theorem S3.2 (Stationary Markov realization fibre of a metric graph triple). *Let $\mathfrak{T}(G, m, c) = (C(V), \mathcal{H}_{m,c}, D_{m,c}, J, \gamma)$ be the metric graph spectral triple of Theorem S2.2. Then:*

(R1) *it has the reversible Markov realization*

$$k_{xy}^{\text{rev}} = \frac{c_{xy}}{m_x}; \quad (83)$$

(R2) every $j \in \mathcal{J}(G, c)$ gives a stationary Markov realization

$$q_{xy} = c_{xy} + j_{xy}, \quad k_{xy} = \frac{c_{xy} + j_{xy}}{m_x}; \quad (84)$$

(R3) conversely, every stationary renewal generator with masses m and symmetric conductances c is uniquely of the form (84);

(R4) every realization in the fibre has the same metric spectral triple.

Thus, up to unread parallel-channel refinements,

$$\text{RReal}_{\text{graph}}(\mathfrak{T}(G, m, c)) \cong \mathcal{J}(G, c). \quad (85)$$

Proof. For (83), the stationary flux is $q_{xy} = c_{xy} = q_{yx}$, so detailed balance holds. For (84), positivity follows from $|j_{xy}| < c_{xy}$. Moreover

$$\sum_y (q_{xy} - q_{yx}) = 2 \sum_y j_{xy} = 0$$

because $Bj = 0$, so m is stationary. Conversely, for a stationary continuous-time Markov generator define $j = (q - q^\top)/2$. Stationarity gives $Bj = 0$, positivity gives the edge inequalities, and $q = c + j$ is unique. The graph spectralization uses only m and $(q + q^\top)/2 = c$. $\square$

Corollary S3.3 (Dimension and faithfulness of the fibre). *For connected G ,*

$$\dim \mathcal{J}(G, c) = |E| - |V| + 1 = b_1(G). \quad (86)$$

The reversible realization is unique exactly when G is a tree. If $b_1(G) > 0$, the metric spectralization map is not faithful.

Proof. The incidence matrix of a connected graph has rank $|V| - 1$, so $\dim \text{Ker } B = |E| - |V| + 1$. The strict inequalities cut out an open neighborhood of zero in this cycle space. A connected graph has zero first Betti number exactly when it is a tree. $\square$

Proposition S3.4 (Entropy production inside one spectral fibre). *For a stationary realization with current j , the steady entropy production is*

$$\sigma(j) = 2 \sum_{e \in E} j_e \log \frac{c_e + j_e}{c_e - j_e} \geq 0. \quad (87)$$

It vanishes exactly at $j = 0$.

Proof. Start from $\frac{1}{2} \sum_{x,y} (q_{xy} - q_{yx}) \log(q_{xy}/q_{yx})$, pair the two orientations of each edge, and substitute $q_e = c_e + j_e$, $q_{\bar{e}} = c_e - j_e$. Each summand is nonnegative because j_e and the logarithm have the same sign; it vanishes only for $j_e = 0$. $\square$

Example S3.5 (Three-cycle universality family). *On an oriented three-cycle with equal conductance c , every constant current $j_e = s$, $|s| < c$, is divergence free. All values of s give one metric spectral triple, whereas*

$$\sigma(s) = 6s \log \frac{c+s}{c-s}$$

varies continuously from zero and diverges as $|s| \uparrow c$.

Theorem S3.6 (Finite spectral universality). *There exist nonisomorphic stationary predictive graph dynamics $T_1 \not\cong T_2$ with isomorphic metric spectral images. More strongly, for every connected graph with $b_1(G) > 0$, the fibre contains a $b_1(G)$ -dimensional open family of dynamically inequivalent renewal laws.*

Proof. Choose two distinct currents in (82). Their directed fluxes and path probabilities differ, and generically their entropy productions differ, so the history-preserving renewal packets are not isomorphic. [Theorem S3.2](#) gives identical metric spectral images, and [Corollary S3.3](#) gives the dimension. $\square$

S3.2 The modular coordinate of the current fibre

For $j \in \mathcal{J}(G, c)$ define the real affinity

$$a_e = \operatorname{arctanh} \frac{j_e}{c_e} = \frac{1}{2} \log \frac{c_e + j_e}{c_e - j_e}. \quad (88)$$

Exact one-cochains are $\operatorname{Ran} B^\top$, so a has a class $[a] \in H^1(G; \mathbb{R}) = \mathbb{R}^E / \operatorname{Ran} B^\top$.

Lemma S3.7 (Zero affinity class forces reversibility). *If $j \in \mathcal{J}(G, c)$ is divergence free and $[a] = 0$, then $j = 0$.*

Proof. Write $a = B^\top \phi$. Since $j_e = c_e \tanh((B^\top \phi)_e)$ and $Bj = 0$,

$$0 = \langle \phi, Bj \rangle = \sum_e c_e (B^\top \phi)_e \tanh((B^\top \phi)_e).$$

Every summand is nonnegative and vanishes only when the edge difference is zero. Connectedness makes ϕ constant, hence $a = j = 0$. $\square$

Theorem S3.8 (Nonlinear Hodge theorem for stationary affinities). *The map*

$$\mathfrak{A} : \mathcal{J}(G, c) \longrightarrow H^1(G; \mathbb{R}), \quad j \longmapsto [\operatorname{arctanh}(j/c)] \quad (89)$$

is a diffeomorphism. More explicitly, for a class α and representative a_0 , define on mean-zero vertex potentials

$$\mathcal{F}_\alpha(\phi) = \sum_{e \in E} c_e \log \cosh(a_{0,e} + (B^\top \phi)_e). \quad (90)$$

This functional is coercive and strictly convex. Its unique minimizer ϕ_α gives

$$a_\alpha = a_0 + B^\top \phi_\alpha, \quad j_{\alpha,e} = c_e \tanh(a_{\alpha,e}), \quad Bj_\alpha = 0, \quad (91)$$

and this is the unique current with modular class α .

Proof. On mean-zero potentials, connectedness gives a Poincaré estimate $\|\phi\| \leq C \|B^\top \phi\|$. Since $\log \cosh t \geq |t| - \log 2$, the functional is coercive. Its Hessian is

$$B \operatorname{diag} \left(c_e \operatorname{sech}^2(a_{0,e} + (B^\top \phi)_e) \right) B^\top,$$

which is positive definite on the mean-zero subspace. Thus there is a unique minimizer. Its Euler equation is

$$0 = \sum_e c_e \tanh(a_{\alpha,e}) (B^\top \psi)_e = \langle Bj_\alpha, \psi \rangle,$$

so j_α is divergence free. Conversely, if j is stationary and $a = \operatorname{arctanh}(j/c)$, then zero is the critical point of (90) in the class $[a]$; strict convexity makes it the unique minimizer. Smooth dependence follows from the implicit-function theorem and the positive Hessian. $\square$

Corollary S3.9 (Faithful metric-modular enrichment). *For fixed (G, m, c) , the enriched object*

$$\mathfrak{S}_{\text{mod}}(T) = (C(V), \mathcal{H}_{m,c}, D_{m,c}, J, \gamma; [a]) \quad (92)$$

determines the complete stationary generator through

$$[a] \mapsto a_\alpha \mapsto j = c \tanh a_\alpha \mapsto q = c + j \mapsto k_{xy} = q_{xy}/m_x.$$

Hence the modular enrichment is faithful on each finite graph fibre.

Proof. Apply [Theorem S3.8](#) and [Theorem S3.2](#). □

The entropy production is the modular pairing

$$\sigma(j) = 4 \sum_{e \in E} j_e a_e, \quad (93)$$

and is unchanged by $a \mapsto a + B^\top \phi$ because $\langle j, B^\top \phi \rangle = \langle Bj, \phi \rangle = 0$.

S3.3 Relative modular data and cellular curvature

Let E^{or} contain both orientations of every edge, with reversal $e \mapsto \bar{e}$, source $s(e)$, terminal $t(e) = s(\bar{e})$, and positive directed stationary flux q_e . On $\mathcal{H}_E = \ell^2(E^{\text{or}})$, define

$$(J_E \xi)(e) = \overline{\xi(\bar{e})}, \quad (\Delta_q \xi)(e) = \frac{q_e}{q_{\bar{e}}} \xi(e) = e^{2a_e} \xi(e), \quad S_q = J_E \Delta_q^{1/2}. \quad (94)$$

Source and terminal multiplication are $(\lambda_E(f)\xi)(e) = f(s(e))\xi(e)$ and $(\rho_E(f)\xi)(e) = f(t(e))\xi(e)$.

Theorem S3.10 (Finite modular-reversal identities). *The packet (94) satisfies*

$$J_E^2 = I, \quad J_E \Delta_q J_E = \Delta_q^{-1}, \quad S_q^2 = I, \quad (95)$$

$$J_E \lambda_E(f^*) J_E = \rho_E(f), \quad S_q \lambda_E(f^*) S_q^{-1} = \rho_E(f), \quad \log \Delta_q(e) = 2a_e. \quad (96)$$

The reversible branch is exactly $\Delta_q = I$.

Proof. Reversal preserves counting measure, so J_E is antiunitary and squares to the identity. Since $a_{\bar{e}} = -a_e$, conjugating the diagonal multiplier e^{2a_e} by J_E gives its inverse; hence $S_q^2 = I$. The source of the reversed edge is the terminal of the original edge, proving the opposite-action identities. The logarithm statement is the definition of a_e . □

For an oriented cycle C ,

$$\text{Hol}_\Delta(C) = \prod_{e \in C} \left(\frac{q_e}{q_{\bar{e}}} \right)^{\epsilon_{C,e}} = \exp \left(2 \sum_{e \in C} \epsilon_{C,e} a_e \right). \quad (97)$$

Now let

$$C^0 \xrightarrow{d_0} C^1 \xrightarrow{d_1} C^2 \xrightarrow{d_2} C^3 \quad (98)$$

be a finite real Hilbert complex and let $\mathcal{H}^1 = \text{Ker } d_1 \cap \text{Ker } d_0^*$.

Theorem S3.11 (Cellular modular Hodge reconstruction). *For $a \in C^1$, define*

$$h_a = P_{\mathcal{H}^1} a, \quad F_a = d_1 a. \quad (99)$$

Then a has the unique orthogonal decomposition

$$a = d_0 \phi + h_a + d_1^* \psi. \quad (100)$$

Its coexact part is reconstructed from the curvature by

$$P_{\text{Ran } d_1^*} a = d_1^* (d_1 d_1^*)^\dagger F_a. \quad (101)$$

Consequently

$$C^1 / \text{Ran } d_0 \longrightarrow \mathcal{H}^1 \oplus \text{Ran } d_1, \quad [a]_{\text{gauge}} \longmapsto (h_a, F_a) \quad (102)$$

is injective and onto the displayed product. A canonical representative is

$$a_{h,F} = h + d_1^* (d_1 d_1^*)^\dagger F. \quad (103)$$

Moreover $d_2 F_a = 0$ and

$$\|a\|^2 = \|d_0 \phi\|^2 + \|h_a\|^2 + \left\langle F_a, (d_1 d_1^*)^\dagger F_a \right\rangle. \quad (104)$$

Proof. Finite Hodge decomposition gives $C^1 = \text{Ran } d_0 \oplus \mathcal{H}^1 \oplus \text{Ran } d_1^*$, proving (100). Since d_1 annihilates the first two summands, $F_a = d_1 d_1^* \psi$; Moore–Penrose inversion on $\text{Ran } d_1$ gives (101). If two cochains have the same (h, F) , their difference has zero harmonic projection and lies in $\text{Ker } d_1$. Its coexact part then has zero norm, so the difference is exact. This proves injectivity; (103) proves surjectivity. Orthogonality gives the norm identity, and $d_2 d_1 = 0$ gives the Bianchi identity. $\square$

Corollary S3.12 (Flat, globally twisted, and curved modular branches). *For a cellular affinity a :*

- (M1) *if $F_a \neq 0$, the modular transport has local face curvature;*
- (M2) *if $F_a = 0$ but $h_a \neq 0$, it is locally flat with nontrivial global positive holonomy;*
- (M3) *if $F_a = h_a = 0$, then $a = d_0 \phi$ and the modular packet is removed by a positive vertex gauge.*

On the periodic three-torus, the global flat sector has dimension three.

Theorem S3.13 (Face holonomy). *For every oriented loaded two-cell p ,*

$$\prod_{e \subset \partial p} \left(\frac{q_e}{q_{\bar{e}}} \right)^{\epsilon_{p,e}} = \exp(2F_a(p)). \quad (105)$$

Thus a source–terminal modular transport descends to a flat cell-line packet exactly when every face-curvature residual vanishes.

Proof. By the cellular coboundary formula, $F_a(p) = \sum_e \epsilon_{p,e} a_e$. Substitute $q_e/q_{\bar{e}} = e^{2a_e}$ and multiply. $\square$

S3.4 Hidden dynamical memory

The graph-current fibre concerns directed dynamics at fixed metric geometry. A second inverse fibre appears when auxiliary memory degrees of freedom have been eliminated into an effective head operator. The dynamic coordinate is the whole resolvent self-energy rather than one static Schur value.

Definition S3.14 (Positive operator-valued tail measure). Let H be finite dimensional. A bounded positive operator-valued tail measure is a weakly countably additive map

$$\mu : \mathfrak{B}((0, \infty)) \rightarrow \mathcal{B}(H), \quad \mu(E) \succeq 0, \quad \|\mu((0, \infty))\| < \infty.$$

Its Stieltjes self-energy is

$$\Sigma_\mu(z) = \int_{(0, \infty)} \frac{1}{\lambda - z} d\mu(\lambda), \quad z \in \mathbb{C} \setminus [0, \infty). \quad (106)$$

Construction S3.15 (Canonical minimal Stieltjes dilation). Let $\mathcal{S}_\mu$ be the finite H -valued Borel simple functions and set

$$\left\langle \sum_j \mathbf{1}_{E_j} \xi_j, \sum_k \mathbf{1}_{F_k} \eta_k \right\rangle_\mu = \sum_{j,k} \langle \xi_j, \mu(E_j \cap F_k) \eta_k \rangle.$$

Quotient by the null space and complete to K_μ . Multiplication by $\mathbf{1}_E$ defines a projection-valued measure $P_\mu(E)$, and

$$C_\mu = \int \lambda dP_\mu(\lambda), \quad V_\mu \xi = [\mathbf{1}_{(0, \infty)} \xi], \quad B_\mu = V_\mu^*.$$

Theorem S3.16 (Operator-measure realization and minimality). *Every bounded positive operator-valued measure satisfies*

$$\Sigma_\mu(z) = B_\mu(C_\mu - z)^{-1} B_\mu^*. \quad (107)$$

Moreover

$$K_\mu = \overline{\text{span}}\{P_\mu(E) B_\mu^* \xi : E \in \mathfrak{B}((0, \infty)), \xi \in H\}. \quad (108)$$

Conversely, if $C \geq 0$ is self-adjoint on K , $B : K \rightarrow H$ is bounded, and

$$K = \overline{\text{span}}\{E_C(E) B^* H\},$$

then $\mu(E) = B E_C(E) B^*$ is a bounded positive operator-valued measure with self-energy $B(C - z)^{-1} B^*$. Two minimal realizations of the same μ are related by a unique unitary U satisfying

$$U C_1 = C_2 U, \quad U B_1^* = B_2^*. \quad (109)$$

Proof. The spectral calculus gives

$$B_\mu(C_\mu - z)^{-1} B_\mu^* = V_\mu^* \left(\int \frac{1}{\lambda - z} dP_\mu(\lambda) \right) V_\mu = \int \frac{1}{\lambda - z} d\mu(\lambda).$$

Every simple function is a finite sum of vectors $P_\mu(E) V_\mu \xi$, proving (108).

For the converse, the spectral theorem gives positivity and countable additivity of $BE_C(E)B^*$. On the algebraic span of spectral-source vectors define

$$U_0 \left(\sum_j E_C(E_j)B^*\xi_j \right) = \sum_j P_\mu(E_j)V_\mu\xi_j.$$

The squared norms on the two sides are both

$$\sum_{j,k} \langle \xi_j, BE_C(E_j \cap E_k)B^*\xi_k \rangle,$$

so U_0 is well defined and isometric. Source minimality makes its extension unitary. It intertwines every spectral projection, hence C , and sends B^* to V_μ . Applying the same construction to two realizations proves existence and uniqueness of (109). $\square$

Theorem S3.17 (Static shadows, inverse moments, and positive Euclidean memory). *For $\text{Im } z > 0$,*

$$\text{Im } \Sigma_\mu(z) = (\text{Im } z) \int \frac{1}{|\lambda - z|^2} d\mu(\lambda) \succeq 0. \quad (110)$$

The finite static shadow

$$Q_0 = \lim_{s \downarrow 0} \Sigma_\mu(-s)$$

exists exactly when the inverse first moment is bounded, and then

$$Q_0 = \int \lambda^{-1} d\mu(\lambda). \quad (111)$$

If the corresponding inverse moment is finite, the r -th boundary jet is

$$\Sigma_{\mu,-}^{(r)}(0) = r! \int \lambda^{-(r+1)} d\mu(\lambda). \quad (112)$$

The Euclidean memory kernel

$$K_\mu(t) = \int e^{-t\lambda} d\mu(\lambda) = B_\mu e^{-tC_\mu} B_\mu^* \quad (113)$$

is positive and

$$\Sigma_\mu(-s) = \int_0^\infty e^{-st} K_\mu(t) dt.$$

Proof. The scalar identity $\text{Im}(\lambda - z)^{-1} = (\text{Im } z)|\lambda - z|^{-2}$ gives the sign after positive operator integration. Monotone convergence along $z = -s$ gives (111). Differentiating $(\lambda - z)^{-1}$ gives the higher inverse moments. The spectral calculus gives the second equality in (113); Tonelli's theorem applied to scalarizations gives the Laplace relation. $\square$

Remark S3.18 (Positive memory is not full OS positivity). Positivity of $K_\mu(t)$ is a two-point positive memory statement. It does not by itself supply a Euclidean field law, a positive-time product algebra, or Osterwalder–Schrader positivity for reflected words.

Theorem S3.19 (Determinacy on controlled support). *If $\text{supp } \mu$ is compact, the complete nonnegative moment sequence $M_n = \int \lambda^n d\mu(\lambda)$ determines μ uniquely. The same conclusion holds on unbounded support if $\int e^{a\lambda} d\text{Tr } \mu(\lambda) < \infty$ for some $a > 0$.*

Proof. On compact support, equality of all moments gives equality against all polynomials; Stone–Weierstrass gives equality against every continuous function, and polarization gives equality of the operator measures.

Under an exponential moment, each scalarized Laplace transform $\int e^{-t\lambda} d\langle \xi, \mu \eta \rangle$ extends holomorphically to a half-plane containing the origin. Its Taylor coefficients are the moments. Equality of all moments therefore gives equality of the Laplace transforms near zero and hence throughout the connected half-plane. Uniqueness of the Laplace transform and polarization recover μ . $\square$

Proposition S3.20 (All integer moments can miss the dynamic self-energy). *For $|\varepsilon| \leq 1$ define*

$$d\mu_\varepsilon(x) = [1 + \varepsilon \sin(2\pi \log x)] \frac{e^{-(\log x)^2/2}}{x\sqrt{2\pi}} dx. \quad (114)$$

Then for every integer $n \in \mathbb{Z}$,

$$\int_0^\infty x^n d\mu_\varepsilon(x) = e^{n^2/2} \quad (115)$$

independently of ε . Nevertheless $\varepsilon \neq \varepsilon'$ implies that the Stieltjes transforms differ for some, hence an open set of, $z \in \mathbb{C} \setminus [0, \infty)$. Consequently all positive moments, all inverse integer moments, and every finite Hankel panel can agree while the dynamic self-energies are inequivalent.

Proof. Set $y = \log x$. The perturbation of the n -th moment is

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{ny - y^2/2} \sin(2\pi y) dy = \operatorname{Im} \exp\left(\frac{(n + 2\pi i)^2}{2}\right) = e^{(n^2 - 4\pi^2)/2} \sin(2\pi n) = 0$$

for every $n \in \mathbb{Z}$. The unperturbed lognormal moment is $e^{n^2/2}$. Distinct ε give distinct measures.

If a finite signed measure on $[0, \infty)$ had identically zero Stieltjes transform on the negative real axis, the identity

$$\int \frac{1}{x + s} d\sigma(x) = \int_0^\infty e^{-st} \left(\int e^{-tx} d\sigma(x) \right) dt$$

and two applications of uniqueness of the Laplace transform would force $\sigma = 0$. Hence distinct measures have distinct Stieltjes transforms. $\square$

Theorem S3.21 (High-energy memory escape and instantaneous contact terms). *Let μ_n be positive tail measures supported in $[m, \infty)$ for one $m > 0$ and define*

$$d\nu_n(\lambda) = \lambda^{-1} d\mu_n(\lambda), \quad Q_n = \nu_n([m, \infty)) = \Sigma_n(0).$$

If $\sup_n \operatorname{Tr} Q_n < \infty$, every sequence has a subsequence and a positive operator-valued measure on the one-point compactification of the form

$$\nu_\infty = \nu_{\text{fin}} + Q_{\text{esc}} \delta_\infty, \quad Q_{\text{esc}} \succeq 0,$$

such that, locally uniformly off $[m, \infty)$,

$$\Sigma_n(z) \longrightarrow Q_{\text{esc}} + \int_{[m, \infty)} \frac{\lambda}{\lambda - z} d\nu_{\text{fin}}(\lambda). \quad (116)$$

The Euclidean memory measures satisfy

$$K_n(t) dt \longrightarrow Q_{\text{esc}} \delta_0 + K_{\text{fin}}(t) dt. \quad (117)$$

Proof. The matrix entries of ν_n are uniformly bounded finite measures on the compact space $[m, \infty]$. Weak-* compactness and polarization give a positive operator-valued limit, which decomposes into finite and point-at-infinity parts. For $z \notin [m, \infty)$, $g_z(\lambda) = \lambda/(\lambda - z)$ extends continuously to infinity with $g_z(\infty) = 1$, proving (116). For $\varphi \in C_c([0, \infty))$,

$$h_\varphi(\lambda) = \int_0^\infty \varphi(t) \lambda e^{-\lambda t} dt$$

extends continuously with $h_\varphi(\infty) = \varphi(0)$; integration against ν_n gives (117). $\square$

Proposition S3.22 (The static triple does not identify escaped memory). *For $Q \succeq 0$ and $\lambda_n \rightarrow \infty$, let*

$$\mu_n = \lambda_n Q \delta_{\lambda_n}.$$

Then

$$\Sigma_n(0) = Q, \quad \Sigma_n(z) = \frac{\lambda_n}{\lambda_n - z} Q \longrightarrow Q, \quad (118)$$

whereas $\lambda_n e^{-\lambda_n t} Q \rightarrow 0$ for every $t > 0$ and converges distributionally to $Q \delta_0$. The same static local correction is therefore compatible with an intrinsic head term or with memory escaping to infinite energy.

Proof. The resolvent identities are immediate, and the exponential kernels form the standard positive approximate identity at the origin. $\square$

S3.5 Global marked monodromy

Definition S3.23 (Marked spectral packet). A marked finite spectral packet is

$$S = (\mathcal{A}, \mathcal{H}, D, J, \gamma; \mathbf{m}), \quad (119)$$

where $\mathbf{m}$ denotes any declared auxiliary data that are required to be preserved by isomorphisms, such as orientation, modular, calibration, frame, or determinant-line markings. Let

$$G_S = \text{Aut}(S) \quad (120)$$

be the compact group of unitary packet automorphisms preserving those markings and the declared algebra automorphism convention.

Theorem S3.24 (Marked spectral-bundle descent). *Let Γ be connected, let each vertex packet be marked-isomorphic to S , and let each oriented edge carry a marked unitary isomorphism U_e with $U_{\bar{e}} = U_e^{-1}$. Then:*

(B1) *after choosing a base vertex and spanning tree, chord holonomies define a homomorphism*

$$\rho_U : \pi_1(\Gamma) \longrightarrow G_S; \quad (121)$$

(B2) *changing vertex frames conjugates ρ_U by one element of G_S ;*

(B3) *one global marked trivialization exists if and only if ρ_U is trivial;*

(B4) *in general the isomorphism class of the flat marked spectral bundle is*

$$[\rho_U] \in \text{Hom}(\pi_1(\Gamma), G_S)/G_S. \quad (122)$$

Since $\pi_1(\Gamma)$ is free of rank $b_1(\Gamma)$, the bundle is determined by $b_1(\Gamma)$ elements of G_S modulo simultaneous conjugation.

Proof. Use the edge maps along a spanning tree to identify every vertex packet with the base packet. Each chord gives the ordered product around its fundamental cycle, an element of G_S . These cycle values extend uniquely to a homomorphism from the free fundamental group. Changing the base frame conjugates every cycle value by the same automorphism. Trivial cycle values make the tree identification path independent and hence give one global marked frame. Conversely, a global frame makes every cycle product telescope. This is precisely the clutching classification (122) on a graph. $\square$

Corollary S3.25 (Nontrivial monodromy is a twisted geometry). *If the local packets are marked-isomorphic and the edge identifications satisfy the declared compatibility relations while $[\rho_U]$ is nontrivial, the resulting object is a flat twisted spectral geometry rather than a globally trivial marked packet.*

Theorem S3.26 (Line characters are incomplete shadows of frame holonomy). *Let $\chi_1, \dots, \chi_m : G_S \rightarrow U(1)$ be determinant, charge, or other complex line characters and let $\epsilon_1, \dots, \epsilon_r : G_S \rightarrow \{\pm 1\}$ be real Pfaffian-sign characters. Trivial frame holonomy implies triviality of all induced line holonomies. The converse holds exactly when*

$$\bigcap_j \text{Ker } \chi_j \cap \bigcap_k \text{Ker } \epsilon_k = \{1\}. \quad (123)$$

Proof. Every character sends the identity to the identity. If (123) holds and all character holonomies are trivial, every cycle value of ρ_U lies in the displayed intersection and is therefore the identity. Conversely, if the intersection contains $g \neq 1$, assign g as monodromy on a one-cycle graph. Every listed character is trivial although the marked bundle is not. $\square$

Proposition S3.27 (A trivial determinant line need not trivialize the packet). *Let a marked packet have scalar algebra, zero Dirac operator, and a multiplicity-two Hilbert carrier so that its marked automorphism group contains $SU(2)$. On a one-cycle graph choose monodromy*

$$g = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}, \quad \theta \notin 2\pi\mathbb{Z}. \quad (124)$$

Then the determinant-line holonomy is one although the marked Hilbert bundle has nontrivial monodromy.

Proof. Every $SU(2)$ element has determinant one, while the displayed matrix is not the identity. Apply Theorems S3.24 and S3.26. $\square$

S3.6 Coarse graining of the principal fibres

Let $\partial_G : \mathbb{R}^E \rightarrow \mathbb{R}^V$ be the oriented incidence map and $\mathcal{Z}(G) = \text{Ker } \partial_G$. A vertex quotient $q : V \twoheadrightarrow \bar{V}$ induces maps Q_V and Q_E by summing vertex and oriented edge coefficients over fibres.

Theorem S3.28 (Current-chain functor). *The incidence maps satisfy*

$$\partial_{\bar{G}} Q_E = Q_V \partial_G, \quad (125)$$

so $Q_E \mathcal{Z}(G) \subseteq \mathcal{Z}(\bar{G})$. If $\bar{c}_{\bar{e}} = \sum_{e \mapsto \bar{e}} c_e$, then $Q_E \text{Curr}(G, c) \subseteq \text{Curr}(\bar{G}, \bar{c})$. When the vertex fibres are connected and the coarse graph retains every inter-fibre edge class, the induced map on cycle spaces is onto and

$$0 \longrightarrow \text{Ker}(Q_E|_{\mathcal{Z}(G)}) \longrightarrow \mathcal{Z}(G) \xrightarrow{Q_E} \mathcal{Z}(\bar{G}) \longrightarrow 0, \quad (126)$$

with

$$\dim \text{Ker}(Q_E|_{\mathcal{Z}(G)}) = b_1(G) - b_1(\bar{G}). \quad (127)$$

The stationary entropy production satisfies

$$\sigma_{\bar{G}}(Q_E j) \leq \sigma_G(j). \quad (128)$$

Proof. For a coarse vertex, the divergence of the summed inter-fibre current equals the sum of fine divergences; internal fine edges cancel in pairs. This is exactly (125). The conductance inequality $|(Q_E j)_{\bar{e}}| \leq \sum_{e \mapsto \bar{e}} |j_e| < \bar{c}_{\bar{e}}$ gives the fibre map. Connected fibres allow every coarse cycle to be lifted and completed inside fibres, proving surjectivity. The dimension formula follows from the Betti-number formula for connected graphs. Finally, on every coarse edge the log-sum inequality for the aggregated forward and reverse fluxes gives entropy contraction; summing over coarse edges yields (128). $\square$

For an effective operator A and positive memory measure μ , write

$$F_{A,\mu}(z) = A - z - \Sigma_{\mu}(z), \quad Q_{>\Lambda} = \int_{(\Lambda, \infty)} \lambda^{-1} d\mu(\lambda), \quad (129)$$

and define

$$\text{RG}_{\Lambda}(A, \mu) = (A - Q_{>\Lambda}, \mu|_{(0, \Lambda]}). \quad (130)$$

Theorem S3.29 (Exact memory RG semigroup). *If $(A_{\Lambda}, \mu_{\Lambda}) = \text{RG}_{\Lambda}(A, \mu)$, then*

$$F_{A_{\Lambda}, \mu_{\Lambda}}(0) = F_{A, \mu}(0). \quad (131)$$

For $|z| \leq r < \Lambda$,

$$\|F_{A, \mu}(z) - F_{A_{\Lambda}, \mu_{\Lambda}}(z)\| \leq \frac{r}{\Lambda - r} \|Q_{>\Lambda}\|. \quad (132)$$

Moreover, if $0 < \Lambda_2 \leq \Lambda_1$,

$$\text{RG}_{\Lambda_2} \circ \text{RG}_{\Lambda_1} = \text{RG}_{\Lambda_2}. \quad (133)$$

Proof. Split the Stieltjes integral at Λ . At $z = 0$ the removed high-energy part is exactly $Q_{>\Lambda}$ and is canceled by the head correction. Away from zero,

$$\frac{1}{\lambda - z} - \frac{1}{\lambda} = \frac{z}{\lambda(\lambda - z)},$$

whose norm is at most $r[(\Lambda - r)\lambda]^{-1}$ for $\lambda > \Lambda$. Integrating gives (132). Two successive cutoffs simply split the same measure into disjoint shells; their inverse-moment head corrections add, proving the semigroup law. $\square$

Let $\rho : \pi_1(\Gamma, v_0) \rightarrow G_S$ be marked monodromy and let a graph contraction $q : \Gamma \rightarrow \bar{\Gamma}$ induce a surjection q_* on fundamental groups.

Theorem S3.30 (Exact marked-monodromy descent). *There is a unique $\bar{\rho} : \pi_1(\bar{\Gamma}) \rightarrow G_S$ satisfying*

$$\rho = \bar{\rho} \circ q_* \quad (134)$$

if and only if

$$\text{Ker } q_* \subseteq \text{Ker } \rho. \quad (135)$$

The conjugacy class $[\bar{\rho}]$ is gauge independent and functorial under further contractions. A determinant or Pfaffian character may satisfy its own descended kernel condition even when (135) fails, so line descent does not imply descent of the full marked packet.

Proof. If (134) holds, every element of $\text{Ker } q_*$ lies in $\text{Ker } \rho$. Conversely, if (135) holds, define $\bar{\rho}(q_*\gamma) = \rho(\gamma)$. The kernel condition makes the definition independent of the chosen lift. Surjectivity of q_* gives uniqueness and the quotient-group universal property gives functoriality. The character statement follows by applying the same argument to a nonfaithful one-dimensional representation of G_S . $\square$

Theorem S3.31 (Functorial principal universality packet). *Collect*

$$\mathfrak{U}(G, c; A, \mu; \rho) = (\text{Curr}(G, c), (A, \mu), [\rho]). \quad (136)$$

For a connected-fibre graph quotient, a memory threshold Λ , and a compatible isometric effective-space compression V , define

$$\mathfrak{U} \mapsto (Q_E \text{Curr}(G, c), (V^*(A - Q_{>\Lambda})V, V^*\mu|_{(0, \Lambda]}V), [\bar{\rho}]). \quad (137)$$

When the monodromy kernel condition holds, this map is canonical and composes strictly for composable graph quotients, nested memory thresholds and compatible head maps. Along it, entropy production cannot increase; the zero-energy Schur complement is preserved before effective-space compression; minimal dilation rank and Pontryagin negative-square index cannot increase; and marked monodromy descends exactly when collapsed cycles lie in its kernel.

Proof. The current and entropy statements are Theorem S3.28; the memory statement and exact composition are Theorem S3.29; the monodromy part is Theorem S3.30. Compression of a positive minimal memory dilation carrier cannot increase source rank. For a generalized-Nevanlinna kernel, head compression replaces each finite kernel Gram by a congruence, which cannot increase the number of negative eigenvalues. Each component is uniquely determined by the corresponding quotient data, so the product map is canonical and composes componentwise. $\square$

S4 Analytic details for compact-spin spectralization

The periodic A_3 construction in Section S2 supplies the explicit metric model. This appendix records the analytic estimates used in the compact-spin convergence criterion. The chartwise construction is verified below; the global theorem additionally assumes the atlas stability conditions of Definition S4.7.

S4.1 A positive Renewal stencil for an arbitrary local inverse metric

Let $d \geq 2$ and put

$$\mathcal{R}_d^+ = \{e_i : 1 \leq i \leq d\} \cup \{e_i + e_j, e_i - e_j : 1 \leq i < j \leq d\}. \quad (138)$$

There are exactly d^2 unoriented directions.

Lemma S4.1 (Positive d^2 -direction decomposition). *Let $A = (A_{ij}) \in \text{Sym}_d(\mathbb{R})$ and suppose that for some $\eta > 0$,*

$$|A_{ij}| < \eta \quad (i \neq j), \quad A_{ii} > (d-1)\eta. \quad (139)$$

Define

$$w_i = A_{ii} - (d-1)\eta, \quad w_{ij}^\pm = \frac{1}{2}(\eta \pm A_{ij}). \quad (140)$$

Then every coefficient is strictly positive and

$$A = \sum_i w_i e_i e_i^\top + \sum_{i < j} \left[w_{ij}^+ (e_i + e_j)(e_i + e_j)^\top + w_{ij}^- (e_i - e_j)(e_i - e_j)^\top \right]. \quad (141)$$

Proof. Positivity follows immediately from (139). In the right-hand side of (141), the (i, j) off-diagonal entry is $w_{ij}^+ - w_{ij}^- = A_{ij}$. Every pair containing i contributes $w_{ij}^+ + w_{ij}^- = \eta$ to the i th diagonal, so the diagonal coefficient is $w_i + (d-1)\eta = A_{ii}$. $\square$

A linear coordinate normalization makes g^{-1} equal to I_d at the centre of a chart. By continuity, after shrinking the chart, the inverse metric remains inside one window (139). Compactness gives a finite good cover by such charts.

Let $\rho = \sqrt{\det g}$ and write the decomposition coefficients of g^{-1} as $w_r(x)$, $r \in \mathcal{R}_d^+$. Put $a_r(x) = \rho(x)w_r(x)$. On $Q_h = h\mathbb{Z}^d/\mathbb{Z}^d$ take

$$m_h(x) = h^d \rho(x), \quad c_h(x, r) = \frac{h^{d-2}}{4} (a_r(x) + a_r(x + hr)). \quad (142)$$

The two directed rates are the conductance divided by the corresponding endpoint mass.

Theorem S4.2 (Reversible coframe Renewal process). *The packet (142) is reversible with stationary mass m_h . Its predictable coordinate bracket is*

$$\mathcal{B}_h(x) = \frac{1}{4\rho(x)} \sum_{r \in \mathcal{R}_d^+} [a_r(x + hr) + 2a_r(x) + a_r(x - hr)] r r^\top, \quad (143)$$

and satisfies

$$\|\mathcal{B}_h - g^{-1}\|_{L^\infty} \leq Ch^2. \quad (144)$$

For every $f \in C^4$,

$$\mathcal{L}_h f = \frac{1}{2} \Delta_g f + O(h^2 \|f\|_{C^4}) \quad (145)$$

uniformly on the chart, and the Dirichlet forms converge to

$$-\langle f_h, \mathcal{L}_h f_h \rangle_{m_h} \longrightarrow \frac{1}{2} \int g^{ij} \partial_i f \partial_j \bar{f} \rho \, dx. \quad (146)$$

Proof. Symmetry of c_h gives detailed balance. The bracket follows by summing the two orientations of every direction through a vertex. Taylor expansion of $a_r(x \pm hr)$ cancels odd terms and, together with (141), gives (144). Expanding $f(x \pm hr)$ to fourth order, the first-order terms cancel and the second-order terms contract with the same tensor decomposition. The derivative of the physical density and of the conductances produces the divergence-form first-order part of Δ_g , yielding (145). Summation by parts gives the finite Dirichlet form; the coefficient convergence and Riemann sums give (146). $\square$

Proposition S4.3 (The metric bracket does not select the spin marking). *The positive bracket determines g^{-1} but not an oriented orthonormal coframe, a spin lift, or a global spin structure. A smooth rotation $R(x) \in \text{SO}(d)$ changes the coframe and spin connection without changing g^{-1} or the scalar Renewal generator, and a nontrivial $\mathbb{Z}_2$ spin holonomy is invisible to every local metric bracket.*

S4.2 Density-symmetric spin Dirac and finite Wilson packet

Let S_d be a complex spin module, let γ_a be Hermitian Clifford matrices, and let E_a^j be the inverse oriented orthonormal coframe. Set

$$c^j = E_a^j \gamma_a, \quad c^j c^k + c^k c^j = 2g^{jk} I, \quad (147)$$

and let Ω_j be the Levi-Civita spin connection. On $L^2(Q, S_d; \rho dx)$,

$$D_g = -i \sum_j c^j (\partial_j + \Omega_j). \quad (148)$$

Lemma S4.4 (Exact density-symmetric Dirac identity). *With $U_\rho \psi = \rho^{1/2} \psi$ and $\nabla_j = \partial_j + \Omega_j$,*

$$U_\rho D_g U_\rho^{-1} = -\frac{i}{2} \sum_{j=1}^d (M_{c^j} \nabla_j + \nabla_j M_{c^j}). \quad (149)$$

Proof. Metric compatibility of Clifford multiplication gives

$$\partial_j c^k + [\Omega_j, c^k] + \Gamma_{j\ell}^k c^\ell = 0.$$

Setting $k = j$ and using $\Gamma_{j\ell}^j = \partial_\ell \log \rho$ gives

$$\partial_j c^j + [\Omega_j, c^j] + c^j \partial_j \log \rho = 0.$$

Conjugating (148) by $\rho^{1/2}$ and applying this contracted identity yields (149). $\square$

Introduce one fixed two-dimensional normal factor and put

$$\widehat{S}_d = S_d \otimes \mathbb{C}^2, \quad \widehat{c}^j = c^j \otimes \sigma_1, \quad \Gamma_\perp = I \otimes \sigma_2, \quad \widehat{\gamma} = I \otimes \sigma_3. \quad (150)$$

Let $T_{j,h}^\Omega$ be exact inverse spin parallel transport followed by a lattice shift and define

$$P_{j,h}^\Omega = \frac{T_{j,h}^\Omega - (T_{j,h}^\Omega)^*}{2ih}, \quad W_h^\Omega = \frac{1}{2h} \sum_{j=1}^d (2I - T_{j,h}^\Omega - (T_{j,h}^\Omega)^*). \quad (151)$$

The local doubled Wilson operator is the density-conjugated version of

$$\widetilde{D}_{g,h}^W = \frac{1}{2} \sum_j (M_{\widehat{c}^j} P_{j,h}^\Omega + P_{j,h}^\Omega M_{\widehat{c}^j}) + \varpi \Gamma_\perp W_h^\Omega. \quad (152)$$

For the local estimates, extend the coefficients smoothly to a fixed periodic coordinate box, preserving uniform ellipticity, and use the density-conjugated ℓ^2 norm with cell weight h^d . Such an

extension is an auxiliary local calculation, not a boundary condition on a chart of M . Let $S_{j,h}$ be the ordinary coordinate shift. In a smooth spin frame, $T_{j,h}^\Omega - S_{j,h} = O(h)$ in operator norm. Define

$$\|u\|_{1,h}^2 = \|u\|_h^2 + \sum_j \|h^{-1}(S_{j,h} - I)u\|_h^2. \quad (153)$$

The analogous covariant norm is uniformly equivalent to this one, since the connection coefficients are bounded. All constants below may depend on the fixed atlas and finitely many coefficient bounds, but not on h .

Theorem S4.5 (Frozen symbol, unique species, and uniform graph estimate). *For fixed $\varpi > 0$, the frozen principal symbol of (152) is*

$$q_{h,x_0}(\theta) = \frac{1}{h} \left[\sum_j \widehat{c}^j(x_0) \sin \theta_j + \varpi \Gamma_\perp \sum_j (1 - \cos \theta_j) \right]. \quad (154)$$

It satisfies

$$q_{h,x_0}(\theta)^2 = \frac{1}{h^2} \left[g^{jk}(x_0) \sin \theta_j \sin \theta_k + \varpi^2 \left(\sum_j (1 - \cos \theta_j) \right)^2 \right] I. \quad (155)$$

Its only zero is $\theta = 0$ on the Brillouin torus. On the fixed smooth periodic extension, and for sections localized in the interior of the original chart, there is a cutoff-independent bound

$$\|u\|_{1,h} \leq C(\|\widetilde{D}_{g,h}^W u\|_h + \|u\|_h). \quad (156)$$

The symmetric placement of coefficients makes the finite operator self-adjoint; the fixed doubled Clifford convention supplies the declared real and grading structures.

Proof. The Clifford relations eliminate the mixed terms in the square. Put $s(\theta) = \sum_j (1 - \cos \theta_j)$ and $\ell_h(\theta) = 2h^{-2}s(\theta)$. Uniform ellipticity and the elementary identity

$$\sum_j \sin^2 \theta_j + s(\theta)^2 = 2s(\theta) + 2 \sum_{i < j} (1 - \cos \theta_i)(1 - \cos \theta_j) \geq 2s(\theta)$$

give constants $0 < c \leq C < \infty$ such that

$$c\ell_h(\theta)I \preceq q_{h,x_0}(\theta)^2 \preceq C\ell_h(\theta)I. \quad (157)$$

For the upper bound use $s \leq 2d$ and $\sum_j \sin^2 \theta_j \leq 2s$. The lower bound controls all forward differences, including modes near the nonzero Brillouin corners. Fourier transformation proves the frozen graph estimate.

To pass to variable coefficients, choose a partition with supports of fixed small diameter r , independent of h . On one support the difference of the variable principal part and its frozen value is bounded by $C\omega_c(r)\|u\|_{1,h} + C_r\|u\|_h$, where ω_c is the modulus of continuity of the coframe coefficients. Derivatives of coefficients and the replacement of ordinary shifts by smooth spin links contribute only the second term. Choose r once so that the first term is absorbed by the frozen estimate. For every fixed smooth cutoff χ ,

$$\|[\widetilde{D}_{g,h}^W, M_{\chi,h}]\| \leq C\|\nabla \chi\|_\infty. \quad (158)$$

Indeed, a shift commutator contains $\chi(x + he_j) - \chi(x) = O(h)$, which cancels the h^{-1} factor in both the Dirac and Wilson differences. Apply the local estimate to χu , sum the squared bounds over a fixed finite partition with $\sum \chi^2 = 1$, and use the same discrete product rule for the forward differences in (153). This proves (156). The fixed localization scale is important: no cutoff derivative diverging as $h \rightarrow 0$ is absorbed into a uniform constant. $\square$

Let $\mathcal{J}_h^0$ be isometric piecewise-constant reconstruction and take $S_h = (\mathcal{J}_h^0)^*$, the cell-average sampling map in the chosen density and spin frame. This avoids a dimension-dependent point-evaluation assumption on Sobolev functions. A piecewise-affine reconstruction I_h on a fixed shape-regular refinement of the coordinate cells satisfies

$$\|I_h u\|_{H^1} \leq C\|u\|_{1,h}, \quad \|I_h u - \mathcal{J}_h^0 u\|_{L^2} \leq Ch\|u\|_{1,h}. \quad (159)$$

These inequalities follow cell by cell from the affine interpolation formula and scaling to a unit cell. Thus it is $I_h u$, not the discontinuous piecewise-constant reconstruction itself, that has a uniform H^1 bound.

Lemma S4.6 (Smooth-core consistency and the Wilson term). *For every smooth doubled spinor supported in the coordinate interior,*

$$\left\| \mathcal{J}_h^0 \tilde{D}_{g,h}^W S_h \psi - U_\rho \hat{D}_g U_\rho^{-1} \psi \right\|_{L^2} \leq Ch\|\psi\|_{H^3}. \quad (160)$$

In particular, on this core,

$$\|W_h^\Omega S_h \psi\|_h \leq Ch\|\psi\|_{H^2}. \quad (161)$$

This is a core estimate, not an operator-norm smallness assertion on the full lattice space.

Proof. Taylor's formula with integral remainder, applied to spin-parallel translated cell averages, gives the covariant centered derivative $-i\nabla_j$ and the second-difference expansion $W_h^\Omega = -(h/2)\sum_j \nabla_j^2$ on a smooth core, with the corresponding Sobolev remainders. Cell averaging and reconstruction add $O(h)$ errors controlled by one further derivative. Symmetric multiplication by the smooth coefficients and Lemma S4.4 give (160); applying the second-difference estimate alone gives (161). $\square$

The two roles of the Wilson term must be distinguished. On smooth sections it vanishes as $h \rightarrow 0$, whereas at nonzero lattice corners its size is of order h^{-1} and removes the spurious low-energy species. It is therefore not a bounded perturbation tending to zero in operator norm. This distinction is also central in rigorous continuum-limit studies of lattice Dirac operators [37]. Moreover, for any bounded sequence u_h and fixed smooth ψ , self-adjointness gives

$$|\langle W_h^\Omega u_h, S_h \psi \rangle_h| = |\langle u_h, W_h^\Omega S_h \psi \rangle_h| \leq Ch\|u_h\|_h\|\psi\|_{H^2}.$$

Hence the regularizer leaves no additional distributional term in the limiting Dirac equation. On a closed manifold the intended self-adjoint realization is the closure of D_g on smooth spinors, with domain $H^1(M; S)$ [34]; no chart-boundary domain is introduced.

S4.3 Atlas compatibility and the global resolvent argument

Local consistency and a positive overlap Gram do not, by themselves, imply global norm-resolvent convergence. The following formulation states the required global approximation properties explicitly, so that the analytic conclusion is separate from a choice of mesh and overlap realization.

Definition S4.7 (Compatible stable spin discretization). Let $\widehat{D} = D_{M,s} \otimes \sigma_1$ on $\mathcal{H} = L^2(M; S \otimes \mathbb{C}^2)$, where M is closed. A compatible stable spin discretization is a sequence of finite Hilbert spaces $\mathcal{H}_h$, self-adjoint operators D_h , and isometries $W_h : \mathcal{H}_h \rightarrow \mathcal{H}$ satisfying:

- (A1) $W_h W_h^* \rightarrow I$ strongly. There are discrete first-order norms $\|\cdot\|_{1,h}$ and reconstructions $I_h : \mathcal{H}_h \rightarrow H^1(M; S \otimes \mathbb{C}^2)$ with

$$\|I_h u\|_{H^1} \leq C \|u\|_{1,h}, \quad \|I_h u - W_h u\| \leq \varepsilon_h \|u\|_{1,h}, \quad \varepsilon_h \rightarrow 0. \quad (162)$$

- (A2) The assembled, rather than merely chartwise, operators satisfy

$$\|u\|_{1,h} \leq C(\|D_h u\|_h + \|u\|_h). \quad (163)$$

- (A3) For every smooth doubled spinor ψ there are samples $S_h \psi$ with

$$W_h S_h \psi \rightarrow \psi, \quad W_h D_h S_h \psi \rightarrow \widehat{D} \psi. \quad (164)$$

The global samples use the transition functions of the fixed spin bundle on shared degrees of freedom, including coordinate and density changes. When real and grading structures are asserted, their finite actions intertwine these reconstructions and converge to the declared continuum actions.

Here is the cancellation that a proposed atlas assembly must respect. Choose fixed smooth real cutoffs χ_α supported in chart interiors with $\sum_\alpha \chi_\alpha^2 = 1$. After transporting local spinors and their density factors to the same bundle, the continuum first-order identity is

$$\sum_\alpha \chi_\alpha \widehat{D} \chi_\alpha = \widehat{D}, \quad \sum_\alpha \chi_\alpha [\widehat{D}, \chi_\alpha] = \frac{1}{2} [\widehat{D}, \sum_\alpha \chi_\alpha^2] = 0. \quad (165)$$

Thus the apparent order-one derivatives of the cutoffs cancel; one must not assert that they individually tend to zero. For a finite-range Wilson operator on a common coordinate mesh the analogous localization identity is

$$\sum_\alpha M_{\chi_\alpha, h} W_h^\Omega M_{\chi_\alpha, h} - W_h^\Omega = \frac{1}{2} \sum_\alpha [M_{\chi_\alpha, h}, [W_h^\Omega, M_{\chi_\alpha, h}]], \quad (166)$$

and its norm is at most $Ch \sum_\alpha \|\nabla \chi_\alpha\|_\infty^2$. Each matrix entry contains two cutoff differences and only one factor h^{-1} . Fixed collars keep all such finite-range stencils away from auxiliary chart boundaries for sufficiently small h .

On a shared geometric link $x \rightarrow y$, spin transition matrices satisfy the exact covariance identity

$$U_{\alpha\beta}(x) T_{xy}^\beta U_{\alpha\beta}(y)^{-1} = T_{xy}^\alpha.$$

It removes frame mismatch on that link. It does not identify different coordinate-grid links or supply a stable interpolation between unrelated grids. For independently meshed charts, such interpolation and its first-order error bounds must be verified in (162)–(164). In particular, positivity of a Gram matrix is not a substitute for the assembled graph estimate (163). The local Wilson estimates above establish the local ingredients; the existence of a global assembly satisfying all these properties is an additional construction requirement.

Theorem S4.8 (Norm-resolvent convergence for compatible stable atlases). *For a compatible stable spin discretization in the sense of Definition S4.7, and every $z \notin \mathbb{R}$,*

$$\left\| W_h(D_h - z)^{-1}W_h^* - (\widehat{D} - z)^{-1} \right\| \longrightarrow 0. \quad (167)$$

The same argument on a fixed smooth periodic coordinate extension gives

$$\left\| \mathcal{J}_h^0(D_{g,h}^W - z)^{-1}(\mathcal{J}_h^0)^* - (\widehat{D}_g - z)^{-1} \right\| \longrightarrow 0. \quad (168)$$

This local statement is about that self-adjoint periodic extension, not an unspecified self-adjoint operator on an open chart. Isolated bounded-energy spectral projections converge, and compatible real and grading structures are retained as specified in Definition S4.7.

Proof. Write $R_h(z) = W_h(D_h - z)^{-1}W_h^*$ and $R(z) = (\widehat{D} - z)^{-1}$. Self-adjointness gives

$$\|(D_h - z)^{-1}\| \leq |\operatorname{Im} z|^{-1}, \quad \|D_h(D_h - z)^{-1}\| \leq 1 + |z| |\operatorname{Im} z|^{-1}. \quad (169)$$

For $\|f\| \leq 1$ put $u_h = (D_h - z)^{-1}W_h^*f$. The global graph estimate bounds $\|u_h\|_{1,h}$ uniformly. Therefore $I_h u_h$ lies in a bounded H^1 set, while $\|W_h u_h - I_h u_h\| \rightarrow 0$ uniformly on this resolvent unit ball. Rellich compactness proves collective compactness of the embedded resolvents along the refining sequence. This step uses the reconstruction estimate and does not assign an H^1 norm to a piecewise-constant function.

For fixed f , let $W_h u_h \rightarrow u$ along a subsequence. Test the discrete resolvent equation against $S_h \psi$, with ψ any smooth doubled spinor. Self-adjointness and (164) pass the equation to $(\widehat{D} - z)u = f$ in the distributional sense. Since smooth spinors are a core for the self-adjoint operator $\widehat{D}$, this places u in its adjoint domain, which is H^1 , and gives $u = R(z)f$. Every convergent subsequence has this limit, so $R_h(z) \rightarrow R(z)$ strongly. Applying the same argument at $\bar{z}$ also gives strong convergence of the adjoints.

For clarity, this last fact is needed for the upgrade to norm convergence. If that convergence failed, choose unit vectors f_h with $\|(R_h(z) - R(z))f_h\| \geq \epsilon$. Passing to subsequences, let $f_h \rightharpoonup f$ and use collective compactness to make $R_h(z)f_h$ converge in norm. For each fixed v , strong adjoint convergence implies

$$\langle v, R_h(z)f_h \rangle = \langle R_h(\bar{z})v, f_h \rangle \longrightarrow \langle R(\bar{z})v, f \rangle.$$

Hence its norm limit is $R(z)f$. Compactness of $R(z)$ also gives $R(z)f_h \rightarrow R(z)f$, a contradiction. This proves (167). On the periodic extension, (159), (156), and (160) provide the same hypotheses. The spectral-projection conclusion is the usual consequence of self-adjoint norm-resolvent convergence for isolated spectral clusters. $\square$

S4.4 Independent all-dimensional metric head

For each chart choose a mesoscopic radius $\rho_h \downarrow 0$ with $h/\rho_h \rightarrow 0$. Connect nearby vertices by endpoint blocks whose edge length is the length of the corresponding short physical curve in g . Let D_h^{met} be the direct sum of these source-terminal blocks.

Theorem S4.9 (Exact finite metric head and geodesic convergence). *For $f \in C(V_h)$,*

$$L_h^{\text{met}}(f) = \|[D_h^{\text{met}}, \pi_h(f)]\| = \max_{\{x,y\} \in E_h} \frac{|f(x) - f(y)|}{\ell_h(x,y)}. \quad (170)$$

The pure-state Connes distance is exactly the graph shortest-path metric. On every closed Riemannian manifold,

$$\sup_{x,y \in V_h} |d_h^{\text{met}}(x,y) - d_g(x,y)| \longrightarrow 0. \quad (171)$$

For smooth f , $L_h^{\text{met}}(f|_{V_h}) \rightarrow \|\nabla_g f\|_\infty$.

Proof. The commutator of one endpoint block has off-diagonal entry $(f(y) - f(x))/\ell_h(x,y)$, so the operator norm gives (170). Telescoping along graph paths gives one inequality in the Connes dual formula; the distance-to-a-fixed-vertex function has seminorm one and gives equality.

Every graph edge has the length of an actual curve, so $d_g \leq d_h^{\text{met}}$. Conversely, partition a minimizing geodesic into pieces of length comparable with ρ_h and replace the partition points by mesh vertices within $O(h)$. Bounded geometry gives

$$d_h^{\text{met}}(x,y) \leq d_g(x,y) + C \left(\rho_h + \frac{h}{\rho_h} \right).$$

Choosing, for example, $\rho_h = h^{1/2}$ proves uniform distance convergence. Taylor expansion along the short edges gives the smooth seminorm convergence. $\square$

For these endpoint metric triples the finite Lip-norm is exactly the ordinary Lipschitz seminorm of the graph shortest-path distance. The metric on $S(C(V_h))$, identified with probability measures on V_h , is therefore the Kantorovich transport distance for d_h^{met} [4]. If p, q are probability measures supported on V_h , the same transport plans give

$$\left| \text{mk}_{L_h^{\text{met}}}(p, q) - W_{1,d_g}(p, q) \right| \leq \sup_{x,y \in V_h} |d_h^{\text{met}}(x,y) - d_g(x,y)|.$$

Here W_{1,d_g} denotes the transport distance between the same measures viewed on M . Unlike the general quadratic graph seminorm, this particular endpoint construction thus yields an immediate comparison on all states, not only on vertices.

Corollary S4.10 (Conditional stable compact-spin approximation). *A closed marked Riemannian spin manifold admitting a compatible stable spin discretization has a finite-dimensional approximation of its doubled spin Dirac operator in generalized norm resolvent, together with convergent independent metric heads. Consequently, essential surjectivity of this particular atlas-based spin construction follows if such compatible stable discretizations are supplied for all target manifolds.*

Proof. Apply Theorem S4.8 to the spin operators and Theorem S4.9 to the independent metric heads. The last assertion records explicitly the additional existence requirement. $\square$

S5 Finite rational indefinite memory

The positive Stieltjes theory in Subsection S3.4 is not the full dynamic image. A finite Krein/Pontryagin tail is instead controlled by the number of negative squares of its generalized Nevanlinna kernel.

Definition S5.1 (Normalized rational Hermitian dynamic function). Let H be finite dimensional and let Q be an $\text{End}(H)$ -valued rational function satisfying

$$Q(\bar{z})^* = Q(z), \quad \lim_{z \rightarrow \infty} Q(z) = 0. \quad (172)$$

Write its Laurent expansion at infinity as

$$Q(z) = - \sum_{n \geq 0} M_n z^{-n-1}, \quad M_n = M_n^*. \quad (173)$$

Let $\mathcal{K}_Q$ be the span of Hankel columns $c_j(h) = (M_{j+r}h)_{r \geq 0}$ and define

$$[c_i(u), c_j(v)]_Q = \langle u, M_{i+j}v \rangle. \quad (174)$$

After quotienting the null space, let $A_Q c_j(h) = c_{j+1}(h)$ and $\Gamma_Q h = c_0(h)$.

Theorem S5.2 (Complete normalized rational generalized-Nevanlinna realization). *The construction above is a minimal finite Pontryagin realization and*

$$Q(z) = \Gamma_Q^+ (A_Q - z)^{-1} \Gamma_Q. \quad (175)$$

Its ordinary dimension is the stabilized coefficient-Hankel rank

$$\dim \mathcal{K}_Q = \text{McM}(Q), \quad (176)$$

and its negative index is the inertia of (174). The Nevanlinna kernel

$$N_Q(z, w) = \frac{Q(z) - Q(w)^*}{z - \bar{w}} \quad (177)$$

has exactly that many negative squares. Every other minimal finite Pontryagin realization of Q is related to this one by a unique source-fixing Pontryagin unitary.

Proof. By construction, the output functional L_Q on the Hankel column model obeys $L_Q A_Q^n \Gamma_Q = M_n$. For large z ,

$$L_Q (A_Q - z)^{-1} \Gamma_Q = - \sum_{n \geq 0} \frac{M_n}{z^{n+1}} = Q(z),$$

and rational continuation proves (175). The power-cyclic span of $\Gamma_Q H$ is all of $\mathcal{K}_Q$ by definition. In finite dimension, the resolvent-source span equals the power-cyclic span: expansion at infinity gives one inclusion and a finite Vandermonde inversion at sufficiently many regular spectral parameters gives the other. Thus the realization is source minimal.

The dimension is the stabilized Hankel rank by construction. Choosing a Hilbert majorant and a congruence which writes the nondegenerate Hankel form as a fundamental symmetry J_Q gives $A_Q^* J_Q = J_Q A_Q$ and $Q(z) = \Gamma_Q^* J_Q (A_Q - z)^{-1} \Gamma_Q$. The standard resolvent identity gives

$$N_Q(z, w) = \Gamma_Q^* J_Q (A_Q - z)^{-1} (A_Q - \bar{w})^{-1} \Gamma_Q.$$

Every finite kernel Gram is therefore a compression of the Pontryagin form, so it has at most its negative index. Source minimality makes the resolvent-source vectors span the whole state space and attains the full index. Finally, if two minimal realizations have the same Q , their complete Hankel moments agree. Mapping $A_Q^j \Gamma_Q h$ to the corresponding cyclic vector in the second realization preserves the indefinite form, is onto, fixes the source, and intertwines the state operators. Uniqueness follows from cyclicity. $\square$

Theorem S5.3 (Pole–Hankel invariants). *For a normalized rational Hermitian Q :*

- (P1) *the principal part at each pole determines the local McMillan degree and all Jordan block sizes through shifted principal-part Hankel ranks;*
- (P2) *at each real pole, the principal-part Hankel Gram determines the complete local Pontryagin inertia;*
- (P3) *every nonreal conjugate pole pair forms a neutral paired root space, and its upper-half-plane local McMillan degree contributes equally to positive and negative index;*
- (P4) *the sum of the local contributions equals the minimum global negative index.*

Proof. At a pole λ , multiply by a sufficiently high power of $(z - \lambda)$ and read the finite principal-part coefficient sequence. The ranks of its shifted Hankel matrices are the controllability/observability ranks of the nilpotent local state operator and their successive differences recover the Jordan chain lengths. For a real pole the local Hankel form is Hermitian and Sylvester inertia is invariant under every minimal state-coordinate change, hence gives the local Pontryagin signature. Hermitian symmetry pairs a nonreal root space with its conjugate; their indefinite form is neutral off-diagonal, so the two signs occur with equal multiplicity. The primary decomposition is Pontryagin orthogonal, and the global index is the sum of local indices. $\square$

A general finite rational Hermitian function has a unique split

$$Q(z) = P(z) + Q_0(z), \quad \lim_{z \rightarrow \infty} Q_0(z) = 0, \quad (178)$$

with a Hermitian polynomial P .

Corollary S5.4 (All finite rational Hermitian dynamics). *The strictly proper part Q_0 is realized by Theorem S5.2. The nonconstant polynomial jet P has a canonical finite infinity-chain/regular descriptor realization. Their primary direct sum gives an equivalence, up to source-fixing descriptor Pontryagin unitary,*

$$\begin{array}{c} \text{finite rational Hermitian dynamic functions} \\ \longleftrightarrow \\ \text{minimal regular Pontryagin descriptor realizations.} \end{array} \quad (179)$$

This includes finite real and nonreal poles, arbitrary finite Jordan chains, and all polynomial jets at infinity.

Proof. The pencil of any regular finite descriptor realization has a primary decomposition at infinity and at finite spectral points. On the finite primary block the transfer function is strictly proper and is minimized by Theorem S5.2. On the generalized zero eigenspace of an anchored descriptor pencil, the resolvent expansion is polynomial and is represented by the canonical nilpotent infinity chain. The direct sum attains both minimum state dimensions and indices. Any second minimal realization splits into the same invariant primary blocks, and the unique source-fixing unitaries on the finite and infinity blocks combine to the unique descriptor equivalence. $\square$

Remark S5.5 (Positive OS memory is the index-zero branch). A positive Stieltjes/OS memory packet has zero negative-square index. A nonzero physical Pontryagin index requires an independently occurring fundamental-symmetry or signed packet. Neither OS reflection, temporal orientation, charge conjugation, nor a determinant/Pfaffian line determines that sign by itself. The unrestricted nonrational infinite-dimensional indefinite essential image is not claimed here.

S6 Completed inverse fibres

S6.1 Joint Gram realizations and the finite stratified fibre

Let $E_1, \dots, E_m$ be finite Hilbert spaces representing minimal marginal realization spaces, normalized so that their marginal Gram operators are the identity. A joint realization is encoded by a block correlation operator

$$\mathcal{K} = [K_{ab}]_{a,b=1}^m, \quad K_{aa} = I_{E_a}, \quad \mathcal{K} \succeq 0. \quad (180)$$

Let Corr_X denote this correlation spectrahedron and let $\mathcal{G}_X$ be the product of the unitary gauge groups of the marginal realization spaces.

Theorem S6.1 (Joint Gram classification). *Every minimal joint realization of the retained marginal Gram blocks is classified, up to a unitary on the common realization space, by a point of $\text{Corr}_X / \mathcal{G}_X$. The support rank of $\mathcal{K}$ is the minimum possible joint realization dimension. A representative $\mathcal{K}$ has residual gauge group exactly $\text{Stab}_{\mathcal{G}_X}(\mathcal{K})$.*

Proof. A joint realization has a Gram factorization $\mathcal{K} = S^*S$, with the identity diagonal expressing the chosen normalization of the marginal blocks. Conversely every positive block operator with identity diagonal admits such a factorization. Restricting the realization space to $\overline{\text{Ran } S}$ gives the minimal realization, whose dimension is $\text{rank } \mathcal{K}$ in the finite case. Two minimal Gram factorizations of the same positive operator are related by a unique unitary on the common range. Independent changes of marginal frames act by block congruence, and the residual action is precisely the stated stabilizer. $\square$

Proposition S6.2 (Pairwise positivity need not give a joint realization). *Pairwise admissible Gram blocks need not assemble into one positive joint Gram operator. For example,*

$$\begin{pmatrix} 1 & -3/4 & -3/4 \\ -3/4 & 1 & -3/4 \\ -3/4 & -3/4 & 1 \end{pmatrix} \quad (181)$$

has every 2×2 principal block positive but has eigenvalue $-1/2$. Thus pairwise compatibility does not imply global Gram compatibility.

Let $\mathcal{J}_X$ be the current fibre, $\mathcal{M}_X$ the complete dynamic-memory fibre over the fixed spectral data, $\mathcal{P}_X$ the retained auxiliary-marking fibre, $\mathcal{C}_X^{\text{cal}}$ the residual calibration class, and $\mathcal{R}_X$ the marked monodromy representation space. We call a finite target *fully reduced* when all null and spectator directions have been quotiented out, the domain, weight, orientation, determinant-line, normalization, and auxiliary-marking conventions are fixed, every retained Gram block is minimal, and a specified finite family of bounded separating functionals (equivalently, finite-rank defect operators) has trivial common kernel on any additional retained direction. This last condition is the standard separation hypothesis that replaces the programme-specific “zero-exhaustion” terminology.

Theorem S6.3 (Finite stratified inverse-fibre exhaustion). *For a completed fully reduced finite target $S^\sharp$,*

$$\text{RReal}_X^{\sharp, \min}(S) \cong \bigsqcup_{[\mathcal{K}] \in \text{Corr}_X / \mathcal{G}_X} \frac{\mathcal{J}_X \times \mathcal{M}_X \times \mathcal{P}_X \times \mathcal{C}_X^{\text{cal}} \times \mathcal{R}_X}{\text{Stab}_{\mathcal{G}_X}(\mathcal{K})}. \quad (182)$$

The factors vary, respectively, directed dynamics, dynamical memory, auxiliary markings, calibration, and global bundle data while preserving the fixed spectral packet. In general the quotient is stratified and not a Cartesian product.

Proof. The separating family in the definition of full reduction rules out any additional independent retained direction after the null quotients have been taken. The graph inverse theorem classifies the directed-current freedom; the positive and Pontryagin/descriptor theorems classify returned dynamic memory; the retained auxiliary markings give $\mathcal{P}_X$; the calibration data give $\mathcal{C}_X^{\text{cal}}$; and marked bundle descent gives $\mathcal{R}_X$. Theorem S6.1 classifies all compatible joint Gram realizations and identifies their residual stabilizers. Equivalence of two fully reduced realizations is therefore simultaneous equivalence in these factors under the common stabilizer, yielding (182). Proposition S6.2 explains why the general fibre cannot be replaced by a product of independently quotiented factors. $\square$

S6.2 Quasilocal and semifinite completion

A completed finite-stage system consists of increasing finite-trace projections $P_n \uparrow I$, fully reduced finite fibres $\mathfrak{F}_n$, bonding maps onto their stable images, tightness of the finite realization and operator-measure data, compatible bundle transport, compatible self-adjoint resolvent data, and a compatible normal semifinite weight. In addition, choose at each stage a family $\mathcal{F}_n$ of bounded separating functionals (or finite-rank defect operators), compatible under restriction, such that the cofinal family $\bigcup_n \mathcal{F}_n$ separates every nonzero additional quasilocal retained direction. Equivalently, the common kernel of this cofinal separating family on the quotient by the declared finite-stage coordinates is zero.

Theorem S6.4 (Completed quasilocal/semifinite inverse-fibre theorem). *For such a completed system, the retained inverse fibre is exactly*

$$\text{RReal}_{\infty}^{\sharp, \min}(\mathcal{S}) \cong \varprojlim_n \left[\bigsqcup_{[\mathcal{K}_n] \in \text{Corr}_n / \mathcal{G}_n} \frac{\mathcal{I}_n \times \mathcal{M}_n \times \mathcal{P}_n \times \mathcal{C}_n^{\text{cal}} \times \mathcal{R}_n}{\text{Stab}(\mathcal{K}_n)} \right]. \quad (183)$$

The inverse limit is taken over the stable images, is nonempty exactly when the stable-image system is nonempty, and contains no additional retained coordinate under the stated separation and tightness hypotheses. On a factor-preserving stratum it may be written as the product of the five component inverse limits and the correlation inverse limit followed by the compatible gauge quotient; without factor-preserving transport the stratified form (183) is essential.

Proof. Surjective restriction to stable images makes the inverse system of nonempty compact finite strata satisfy the finite-intersection property, so compatible threads exist whenever every stable image is nonempty. Tightness prevents a nonzero realization direction or dynamic mass from escaping every finite projection. The compatible resolvents determine one closed self-adjoint operator through the resolvent identity, adjoint symmetry, injectivity, and dense range. The compatible finite-trace weights increase to a unique lower-semicontinuous normal semifinite weight by monotone convergence, while compatible bundle transport fixes the remaining global data. Finally, if an additional retained quasilocal coordinate existed, the cofinal separating family would contain a functional or defect operator nonzero on it, contradicting the trivial-common-kernel hypothesis. The finite fibres are given by Theorem S6.3; taking compatible threads proves (183). $\square$